

Quartic Digamma Bridges, Harmonic Tail Moments, and Complex Resolvent Powers

Complete proofs, endpoint corrections,
and normalized antiderivatives

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Abstract

We resolve three concrete parts of the current ProveIt research agenda. First, we evaluate the coordinate finite part of the fourth power of the digamma function through the linear Laurent coefficient of the strict depth-three harmonic zeta function, with every lower-depth and endpoint correction explicit. A normally convergent global partial-fraction identity supplies a normalized primitive and an all-order polygamma hierarchy. Second, we evaluate every centered nonpositive even moment of a product of two even harmonic tails by a finite Bernoulli formula. For each pair of tail orders the entire moment sequence belongs to one fixed span of ordinary zeta values and products, and its convergent generating function reduces to finitely many classical polylogarithms. The first mixed pair is generated rationally by its first four moments. Third, we extend the cotangent resolvent transform to arbitrary complex powers and complete its joint Stieltjes germ at every integer power. The completion includes a finite endpoint term that remains nonzero when the apparent pole is removable. It gives all logarithmic powers, exact parameter derivatives, and normalized primitives. An independent Gamma–Gauss generator evaluates all ordinary polynomial moments. We separate these proofs from computational diagnostics, record the source audit, and identify the remaining arithmetic and higher-depth questions.

Fixed source baseline. ProveIt revision 7bd45777a8a1, including all eleven incoming research archives. The precise source map and full commit identifier appear in Section 1. The Gaussian S_6 and revised S_8 conjectures remain open.

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1 Source baseline and mathematical contributions

This article continues the polylogarithm manuscript and incoming research reports in the ProveIt repository [1] at the fixed revision

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The source comparison uses the manuscript, the relevant archived continuations, and all eleven incoming ZIP archives present at this revision. The incoming directory is a research staging area: a claim being present there is not evidence that it is proved or integrated into a canonical manuscript chapter. We therefore trace each advance to a named question and preserve the qualifications in its source.

The immediate starting points are the cubic harmonic bridge [2], the centered parity and rational-resolvent theorems [3], and the higher-reflection generators [4]. The strict ordered-Hurwitz conventions come from the all-depth germ report [5]. The exact-relation obstruction and the incomplete larger relation search [6, 7] delimit what can presently be claimed about the Gaussian polylogarithm conjectures.

1.1 Three resolved research directions

The main advances are the following. They are new relative to the inspected source corpus; no exhaustive claim of priority across the literature is intended.

1. **The explicitly requested quartic harmonic reduction.** Question 6 of the cubic harmonic report asks for the finite reduction at $k = 4$, with the higher-depth data identified. We prove Theorem 3.1, fixing the entire remainder in the quartic Mittag-Leffler expansion and all integration constants. The answer uses the linear Laurent coefficient θ of $Z_3(1 + u, 1, 1)$, the inherited coefficient η of $Z_2(1 + u, 1)$, and ordinary zeta and Stieltjes constants. Equation (3.5) eliminates η by combining the fourth and third moments. An absolutely convergent harmonic series, an elementary Mellin kernel for θ , a normalized primitive, and every argument derivative accompany the evaluation.
2. **Every moment of every pair of even harmonic tails.** Question R1 of the harmonic-parity report asks first for the mixed pair of orders two and four, and then for a controlled algebra at general orders. Theorems 4.2 and 5.1 solve the pure two-tail family for all even $p, q \geq 2$ and all moment orders. The finite formula retains no unreduced double-zeta coordinate: all double zeta values have odd weight and reduce by classical identities. The moment order does not enlarge the specified arithmetic span. For $(p, q) = (2, 4)$, every moment is a rational combination of the first four. The additional factors of H_n and odd-order harmonic numbers mentioned in R1 remain outside this theorem.
3. **Complex powers and full moving-endpoint completion.** Question R4 of the same report asks for a branch, a joint meromorphic generator, and the full resonant amplitudes for $(\pi \cot \pi x - z)^{-\lambda}$. Theorems 6.1, 6.2, and 6.4 provide them. The result applies to every nonreal z , every Stieltjes index and argument derivative, and every integer power with arbitrary logarithmic powers. Away from integer powers the uncompleted local formula applies directly. The correction at negative integers is finite even when its denominator is removable; omitting it changes the answer. Section 7 adds a Gamma-Gauss generator for all ordinary polynomial moments and a generalized-harmonic formula at the imaginary base points.

The status changes are deliberately narrower than the broadest questions in the sources. The quartic calculation does not settle every power $k \geq 4$. The two-tail theorem does not treat four

tails or extra harmonic factors. A functional transform for arbitrary complex powers need not reduce every one of its parameter values to a finite list of familiar constants. Each theorem states exactly what is finite, what is continued, and what arithmetic coordinate remains.

1.2 Representative identities

Two compact consequences illustrate the range of the results. With $x_n = n + 1/2$, one obtains the ordinary absolutely convergent sum

$$\sum_{n=0}^{\infty} \left\{ \frac{1344x_n^6 + 2000x_n^4 + 412x_n^2 + 7}{6} \psi'(n+1)\psi'''(n+1) - 448x_n^2 - \frac{1216}{3} \right\} = -88. \quad (1.1)$$

The polynomial subtraction is part of the summand; it is not a separate divergent series. The proof is exact elimination among four independently specified centered moments, as detailed in (4.26)–(4.27).

At a negative half power of the resolvent, an ordinary real integral has the elementary zeta/logarithm evaluation

$$\int_0^1 \left(x^2 - x + \frac{1}{6} \right) \frac{\cos(\pi x/2)}{\sqrt{\sin(\pi x)}} dx = \frac{1}{12} - \frac{\log^2 2}{\pi^2}. \quad (1.2)$$

Its entire sequence of Bernoulli analogues follows from one Gamma quotient, rather than separate evaluations of successive moments.

These ordinary identities complement the finite-part identity for ψ^4 . A finite part is appropriate in that case because the unsubtracted integral diverges at zero; it is not required for (1.1) or (1.2).

1.3 Classical inputs and proof architecture

The classical ingredients are digamma recurrence and reflection, Hurwitz Fourier expansions, Mellin transforms, Bernoulli polynomials, Euler sums, and beta/hypergeometric integral formulas [8, 9, 10]. Harmonic-zeta and Stieltjes coordinates have their own established literature; Coppo's harmonic-zeta identities provide a recent point of comparison [11]. We use these sources for specified formulas, not to infer any unproved constant reduction.

Three proof mechanisms perform different jobs. A two-point contour fixes the entire function that remains after matching the digamma principal parts. Exact finite summation by parts identifies the constant at infinity in a harmonic-tail moment without rearranging a divergent double sum. Finally, local monomial subtraction in two complex parameters distinguishes a meromorphic spectral value from the coordinate finite part of its specialized integrand. The last distinction is essential for differentiation and antidifferentiation at resonances.

The proofs do not depend on a numerical relation search. The accompanying scripts check finite algebra independently and compare analytically different numerical representations. Explicit truncation estimates support these comparisons, but the recorded floating-point runs do not constitute interval certificates or formal proof-assistant proofs.

2 Conventions and finite-part normalization

We use $\psi = \Gamma'/\Gamma$, $\psi^{(0)} = \psi$, and $H_n^{(r)} = \sum_{j=1}^n j^{-r}$, with $H_n = H_n^{(1)}$ and $H_0^{(r)} = 0$. A prime on $\zeta(s)$ denotes differentiation in its order s ; a superscript $[k]$ on Li_s also denotes order differentiation. Derivatives of $\gamma_m(x)$ are with respect to its argument x .

The generalized Stieltjes convention is

$$\zeta(s, x) = \frac{1}{s-1} + \sum_{m \geq 0} \frac{(-1)^m \gamma_m(x)}{m!} (s-1)^m. \quad (2.1)$$

Thus $\gamma_0(x) = -\psi(x)$ and $\gamma_m = \gamma_m(1)$. After setting $s = 1 - u$, the pole-removed germ is

$$\zeta(1-u, x) + \frac{1}{u} = \sum_{m \geq 0} \gamma_m(x) \frac{u^m}{m!}.$$

This positive-sign u expansion must not be confused with the alternating-sign expansion of $\zeta(1+u)$.

Strict multiple zeta functions have decreasing indices:

$$Z_r(s_1, \dots, s_r) = \sum_{n_1 > \dots > n_r \geq 1} \prod_{j=1}^r n_j^{-s_j}.$$

We also write $\zeta(a, b) = Z_2(a, b)$ when the sum converges. The coefficients η and θ in Section 3 are coefficients of u^1 , with no hidden factorial or sign. Bernoulli numbers have $B_1 = -1/2$, and Bernoulli polynomials use

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} B_n(x) \frac{t^n}{n!}.$$

Definition 2.1 (Coordinate finite part). Suppose an integrand on $(0, 1)$ has, to every required finite order, endpoint expansions in complex powers and powers of logarithms, with an integrable remainder after finitely many subtractions. Its coordinate finite part is the constant coefficient of the cutoff integral in the endpoint coordinates x and $1 - x$. One may introduce independent cutoffs at the two ends; the constant is the coefficient of both cutoff powers zero and both logarithm powers zero. A common cutoff gives the same constant for the separate endpoint expansions used here.

For example,

$$\text{FP} \int_0^1 x^{-j} dx = -\frac{1}{j-1} \quad (j \geq 2), \quad \text{FP} \int_0^1 \frac{\log^k x}{x} dx = 0 \quad (k \geq 0).$$

At a general nonzero exponent α the regularized monomial integral is $1/\alpha$; its parameter derivatives are the corresponding logarithmic integrals. At $\alpha = 0$ the coordinate constant is zero, whereas the meromorphic function has a pole. This elementary difference becomes a finite contact term when a coefficient also vanishes at the same parameter.

A change of endpoint coordinate can change a finite part. Every identity in this article fixes the coordinate before taking the constant, differentiating parameters, or forming a primitive. In particular, the spectral values $D_{p,q}(-2r)$ are initially defined by meromorphic continuation, and their equality to ordinary subtracted sums is proved in Proposition 4.1; it is not an implicit choice of summation method.

All primitives are normalized. The quartic primitive has zero constant term at its singular origin. Primitives in the nonreal resolvent parameter are differences at a specified base point within one half-plane. Complex powers of the resolvent use the continuous logarithm specified in Section 6; no branch is inferred from an unqualified power of a complex number.

3 The quartic digamma finite part and a depth-three harmonic coefficient

The cubic harmonic report [2] leaves a specific next question: carry out the power-to-harmonic reduction for the fourth power of the digamma function, with the entire remainder and every endpoint constant fixed. The theorem below completes that reduction. The retained depth-three coefficient is defined explicitly; a finite evaluation of it in ordinary zeta and Stieltjes constants is a separate arithmetic question.

Throughout this section, write $\gamma = \gamma_0$, and use

$$\zeta(1+u) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j}{j!} u^j.$$

For strict multiple zeta functions the larger index is first:

$$Z_r(s_1, \dots, s_r) = \sum_{n_1 > \dots > n_r \geq 1} n_1^{-s_1} \dots n_r^{-s_r}.$$

Only a single spectral order varies in the following definitions:

$$Z_2(1+u, 1) = u^{-2} + \gamma u^{-1} + \frac{\gamma^2 - \zeta(2)}{2} + \eta u + O(u^2), \quad (3.1)$$

$$Z_3(1+u, 1, 1) = u^{-3} + \gamma u^{-2} + \frac{\gamma^2 - \zeta(2)}{2u} + \frac{\gamma^3 - 3\gamma\zeta(2) + 2\zeta(3)}{6} + \theta u + O(u^2). \quad (3.2)$$

Thus θ is the linear Laurent coefficient, with no factorial or sign included. These conventions agree with the incoming ordered-Hurwitz and cubic harmonic reports [5, 2]. The expansions can also be established directly from the finite elementary symmetric sums used below.

Define $\text{FP} \int_0^1$ as the constant coefficient of $\int_\varepsilon^1$ in the coordinate ε itself. In particular, $\text{FP} \int_0^1 x^{-j} dx = -1/(j-1)$ for $j \geq 2$, whereas $\text{FP} \int_0^1 x^{-1} dx = 0$.

Theorem 3.1 (Quartic harmonic bridge). *Let*

$$A_4 = 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4) + 4\zeta'(3).$$

Then

$$Q_4 := \text{FP} \int_0^1 \psi(x)^4 dx = A_4 - 24\theta + 24\gamma\eta + 12(\gamma^2 - \zeta(2))\gamma_1. \quad (3.3)$$

Equivalently, an absolutely convergent arithmetic representation is

$$Q_4 = A_4 + 12\gamma_3 - 24\zeta(2)\gamma_1 + 12 \sum_{n=1}^{\infty} \frac{(\psi(n)^2 + \psi'(n) - \log^2 n) \log n}{n}. \quad (3.4)$$

In generalized harmonic notation the numerator inside the parentheses is

$$(H_{n-1} - \gamma)^2 + \zeta(2) - H_{n-1}^{(2)} - \log^2 n.$$

Using the already proved cubic bridge, one may isolate the single depth-three coefficient in

$$\begin{aligned} \text{FP} \int_0^1 \{\psi(x)^4 + 4\gamma\psi(x)^3\} dx &= 12\zeta(3) + 18\zeta(4) + 4\zeta'(3) \\ &\quad - 24\theta - 12(\gamma^2 + \zeta(2))\gamma_1. \end{aligned} \quad (3.5)$$

No arithmetic independence or ordinary-constant evaluation of θ is asserted by these formulas.

3.1 A global quartic partial-fraction identity

For $n \geq 0$ put

$$\begin{aligned} b_n &= H_n - \gamma, & c_n &= \zeta(2) + H_n^{(2)}, & d_n &= H_n^{(3)} - \zeta(3), \\ q_n &= 6b_n^2 - 4c_n, & a_n &= -b_n^3 + 3b_nc_n - d_n, \end{aligned} \quad (3.6)$$

with $H_0^{(r)} = 0$. The quartic principal part at $-n$ is

$$P_n(z) = \frac{1}{(z+n)^4} - \frac{4b_n}{(z+n)^3} + \frac{q_n}{(z+n)^2} + \frac{4a_n}{z+n}. \quad (3.7)$$

The constant coefficient of $\psi(z)^4$ at $z = 0$ is

$$\kappa_4 = \gamma^4 - 12\gamma^2\zeta(2) + 6\zeta(2)^2 + 12\gamma\zeta(3) - 4\zeta(4). \quad (3.8)$$

Lemma 3.2 (The entire remainder is fixed). *For $z \notin \{0, -1, -2, \dots\}$,*

$$\psi(z)^4 = P_0(z) + \kappa_4 + \sum_{n=1}^{\infty} (P_n(z) - P_n(0)). \quad (3.9)$$

The series converges normally on every compact set avoiding the poles.

Proof. The digamma recurrence and the Taylor expansion at zero [8, Sections 5.5, 5.7] give

$$\psi(-n + \varepsilon) = -\varepsilon^{-1} + b_n + c_n\varepsilon + d_n\varepsilon^2 + (\zeta(4) + H_n^{(4)})\varepsilon^3 + O(\varepsilon^4).$$

Taking its fourth power proves (3.7); its constant term at zero is (3.8). Matching principal parts by itself would leave an unknown entire function, so one additional argument is necessary.

Let $f(z) = \psi(z)^4 - P_0(z)$, which is regular at zero. For fixed z , integrate

$$\frac{zf(w)}{w(w-z)}$$

over $|w| = N + 1/2$, with N sufficiently large. On these circles, $f(w) = O(\log^4 N)$ uniformly. In the right half-plane this follows from the usual digamma asymptotic [8, Section 5.11]. In the left half-plane use $\psi(w) = \psi(1-w) - \pi \cot(\pi w)$. The cotangent is bounded uniformly on the chosen circles: for $|\operatorname{Im} w| \geq 1/4$ its exponential formula suffices, and for the remaining arcs the real part remains a fixed distance from every integer. Hence the contour integral is $O(\log^4 N/N)$ and tends to zero.

Its residues at z and 0 are $f(z)$ and $-f(0)$, and the residue at $-n$ is $-(P_n(z) - P_n(0))$. The residue theorem proves (3.9). Finally, $b_n = O(\log n)$, $q_n = O(\log^2 n)$ and $a_n = O(\log^3 n)$, while $(n+z)^{-1} - n^{-1} = O(n^{-2})$ uniformly on a fixed compact set. This proves normal convergence as well as the identity. $\square$

3.2 Integration and finite summation by parts

Integrating the normally convergent remainder in (3.9) term by term gives

$$\begin{aligned} Q_4 &= -\frac{1}{3} - 2\gamma - q_0 + \kappa_4 \\ &\quad + \sum_{n \geq 1} \left\{ \frac{n^{-3} - (n+1)^{-3}}{3} - n^{-4} - 2b_n(n^{-2} - (n+1)^{-2}) + 4b_nn^{-3} \right. \\ &\quad \left. + q_n(n^{-1} - (n+1)^{-1} - n^{-2}) + 4a_n \left(\log \left(1 + \frac{1}{n} \right) - \frac{1}{n} \right) \right\}. \end{aligned} \quad (3.10)$$

The finite parts $-1/3$, -2γ and $-q_0$ are retained explicitly. The simple-pole finite part at the origin is zero.

We use the following classical convergent Euler sums [8, Equations 25.16.8 and 25.16.13]:

$$\begin{aligned} \sum_{n \geq 1} \frac{H_n}{n^2} &= 2\zeta(3), & \sum_{n \geq 1} \frac{H_n}{n^3} &= \frac{5}{4}\zeta(4), \\ \sum_{n \geq 1} \frac{H_n^2}{n^2} &= \frac{17}{4}\zeta(4), & \sum_{n \geq 1} \frac{H_n^{(2)}}{n^2} &= \frac{\zeta(2)^2 + \zeta(4)}{2}. \end{aligned}$$

The first three can be taken from the standard Euler-sum formulas; the fourth follows by splitting the product $\zeta(2)^2$ into its two strict triangles and its diagonal. Also $\zeta(2)^2 = 5\zeta(4)/2$. The exact differences

$$b_n - b_{n-1} = \frac{1}{n}, \quad q_n - q_{n-1} = \frac{12b_{n-1}}{n} + \frac{2}{n^2}, \quad a_n - a_{n-1} = \frac{3(c_{n-1} - b_{n-1}^2)}{n} + \frac{1}{n^3} \quad (3.11)$$

then reduce all terms in (3.10) except the logarithmic ones to

$$Q_4 = B_4 + 4 \sum_{n \geq 1} a_n \left(\log \left(1 + \frac{1}{n} \right) - \frac{1}{n} \right), \quad (3.12)$$

where

$$B_4 = \gamma^4 - 18\gamma^2\zeta(2) - 12\gamma\zeta(2) + 32\gamma\zeta(3) + 12\zeta(3) + \frac{13}{2}\zeta(4).$$

For a directly checkable bookkeeping formula, the lower-pole terms before substitution of the Euler sums are

$$\kappa_4 - \zeta(4) + 4 \sum_{n \geq 1} \frac{b_n}{n^3} + 12 \sum_{n \geq 1} \frac{b_{n-1}}{n^2} - 6 \sum_{n \geq 1} \frac{b_n^2}{n^2} + 4 \sum_{n \geq 1} \frac{c_n}{n^2}.$$

This equals B_4 exactly.

The residue difference in (3.11) is the decisive step. For a finite integer $N \geq 1$, define

$$M_N = \sum_{n=1}^N \frac{(b_{n-1}^2 - c_{n-1}) \log n}{n}.$$

Finite summation by parts gives the exact equality

$$\sum_{n=1}^N a_n \log \left(1 + \frac{1}{n} \right) = a_N \log(N+1) + 3M_N - \sum_{n=1}^N \frac{\log n}{n^3}. \quad (3.13)$$

To evaluate the other finite sum, let $e_j(N)$ be the elementary symmetric sum of degree j in $1, 1/2, \dots, 1/N$, and write

$$H_N(r_1, \dots, r_d) = \sum_{N \geq n_1 > \dots > n_d \geq 1} n_1^{-r_1} \dots n_d^{-r_d}.$$

The polynomial expression for the residue is

$$a_n = -6e_3(n) + 6\gamma e_2(n) + 3(\zeta(2) - \gamma^2)H_n + \gamma^3 - 3\gamma\zeta(2) + \zeta(3) + H_n^{(3)}. \quad (3.14)$$

Use $e_j(n) = e_j(n-1) + e_{j-1}(n-1)/n$ in each sum, and split the product $H_N H_N^{(3)}$ into strict triangles. This proves, without any infinite rearrangement,

$$\begin{aligned} \sum_{n=1}^N \frac{a_n}{n} &= -6e_4(N) + 6\gamma e_3(N) + 3(\zeta(2) - \gamma^2)e_2(N) \\ &\quad + (\gamma^3 - 3\gamma\zeta(2) + \zeta(3))H_N + H_N H_N^{(3)} \\ &\quad - 6H_N(2, 1, 1) + 6\gamma H_N(2, 1) + 3(\zeta(2) - \gamma^2)H_N^{(2)} - H_N(3, 1). \end{aligned} \quad (3.15)$$

The only multiple-zeta limits needed here are

$$Z_3(2, 1, 1) = \zeta(4), \quad Z_2(2, 1) = \zeta(3), \quad Z_2(3, 1) = \frac{1}{4}\zeta(4).$$

For example,

$$Z_3(2, 1, 1) = \frac{1}{6} \int_0^1 \frac{[-\log(1-t)]^3}{t} dt = \frac{1}{6} \int_0^1 \frac{(-\log t)^3}{1-t} dt = \zeta(4),$$

and the double-zeta values follow from the displayed Euler sums.

The elementary symmetric formulas are

$$\begin{aligned} e_2(N) &= \frac{H_N^2 - H_N^{(2)}}{2}, \\ e_3(N) &= \frac{H_N^3 - 3H_N H_N^{(2)} + 2H_N^{(3)}}{6}, \\ e_4(N) &= \frac{H_N^4 - 6H_N^2 H_N^{(2)} + 3(H_N^{(2)})^2 + 8H_N H_N^{(3)} - 6H_N^{(4)}}{24}. \end{aligned}$$

Substitute these into (3.15), using $H_N = \log N + \gamma + O(N^{-1})$ and $H_N^{(r)} = \zeta(r) + O(N^{1-r})$ for $r \geq 2$. All discarded terms are $o(1)$, including their polynomial logarithmic factors. The result, including its full constant term, is

$$\sum_{n=1}^N \frac{a_n}{n} = -\frac{\log^4 N}{4} + 3\zeta(2) \log^2 N + C_4 + o(1), \quad (3.16)$$

where

$$C_4 = \frac{\gamma^4}{4} - \frac{9}{2}\gamma^2\zeta(2) + 8\gamma\zeta(3) - \frac{23}{8}\zeta(4). \quad (3.17)$$

Also

$$a_N \log(N+1) = -\log^4 N + 6\zeta(2) \log^2 N + o(1).$$

Equations (3.12)–(3.16) now yield

$$Q_4 = A_4 + 12 \lim_{N \rightarrow \infty} \left\{ M_N - \frac{\log^4 N}{4} + \zeta(2) \log^2 N \right\}. \quad (3.18)$$

Indeed $B_4 - 4C_4 = 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4)$, and $-\sum \log n/n^3 = \zeta'(3)$. This is where an omitted endpoint constant or a sign error in the logarithmic sum would alter the theorem.

3.3 Identification of the Laurent coefficient

Since

$$b_{n-1}^2 - c_{n-1} = \psi(n)^2 + \psi'(n) - 2\zeta(2),$$

and $\psi(n) = \log n + O(n^{-1})$, $\psi'(n) = O(n^{-1})$, the series in (3.4) is absolutely convergent, with summands $O(\log^2 n/n^2)$. The standard Stieltjes limits give

$$\begin{aligned} \lim_{N \rightarrow \infty} \left\{ M_N - \frac{\log^4 N}{4} + \zeta(2) \log^2 N \right\} &= \gamma_3 - 2\zeta(2)\gamma_1 \\ &+ \sum_{n \geq 1} \frac{(\psi(n)^2 + \psi'(n) - \log^2 n) \log n}{n}. \end{aligned} \quad (3.19)$$

Together with (3.18), this proves (3.4).

For the precise spectral interpretation, define the holomorphic function

$$D_4(s) = \sum_{n \geq 1} \frac{\psi(n)^2 + \psi'(n) - \log^2 n}{n^s}, \quad \operatorname{Re} s > 0.$$

The series and every fixed derivative converge normally on compact subsets of this half-plane. In $\operatorname{Re} s > 1$ the finite identity $H_{n-1}^2 - H_{n-1}^{(2)} = 2e_2(n-1)$ gives

$$D_4(s) = 2Z_3(s, 1, 1) - 2\gamma Z_2(s, 1) + (\gamma^2 + \zeta(2))\zeta(s) - \zeta''(s). \quad (3.20)$$

Analytic continuation carries the identity to the germ at $s = 1$. Comparing its linear coefficients, with (3.1) and (3.2), yields

$$D_4'(1) = 2\theta - 2\gamma\eta - (\gamma^2 + \zeta(2))\gamma_1 + \gamma_3.$$

The convergent sum in (3.4) is $-D_4'(1)$. Substitution proves (3.3). Finally, the incoming cubic report [2] proves

$$Q_3 = \text{FP} \int_0^1 \psi(x)^3 dx = 3\zeta(2) - 6\eta - 6\gamma\gamma_1.$$

Adding $4\gamma Q_3$ gives (3.5).

A direct existence check for the depth-three Laurent germ. There is also a useful elementary Mellin representation:

$$Z_3(s, 1, 1) = \frac{1}{2\Gamma(s)} \int_0^\infty \frac{t^{s-1} \log^2(1 - e^{-t})}{e^t - 1} dt, \quad \operatorname{Re} s > 1. \quad (3.21)$$

Indeed the generating series of $e_2(n-1)$ is $\log^2(1-z)/(2(1/z-1))$. Mellin integration is absolutely convergent in the stated domain. Near zero the kernel is $\log^2 t/(2t) + O(\log^2 t)$, so subtracting that one local term gives a holomorphic remainder $R_3(u)$ in

$$\Gamma(1+u)Z_3(1+u, 1, 1) = u^{-3} + R_3(u).$$

The identity $\frac{d}{dt} \log^3(1 - e^{-t}) = 3\log^2(1 - e^{-t})/(e^t - 1)$ shows $R_3(0) = 0$. The inverse-Gamma expansion therefore gives exactly the principal and constant coefficients in (3.2). In particular,

$$\begin{aligned} \theta &= \frac{\gamma^4 - 6\gamma^2\zeta(2) + 3\zeta(2)^2 + 8\gamma\zeta(3) - 6\zeta(4)}{24} \\ &\quad + \frac{1}{2} \int_0^1 \log t \left\{ \frac{\log^2(1 - e^{-t})}{e^t - 1} - \frac{\log^2 t}{t} \right\} dt \\ &\quad + \frac{1}{2} \int_1^\infty \frac{\log t \log^2(1 - e^{-t})}{e^t - 1} dt. \end{aligned} \quad (3.22)$$

Both integrals in this display converge ordinarily. This fixes the retained coordinate by an elementary, one-dimensional kernel as well as by the strict Laurent definition; it does not remove its arithmetic content by renaming it.

3.4 A normalized quartic antiderivative and derivative identities

For $j \geq 2$, define

$$F_j(n, z) = \frac{(n+z)^{1-j} - n^{1-j}}{1-j} - \frac{z}{n^j}, \quad F_1(n, z) = \log\left(1 + \frac{z}{n}\right) - \frac{z}{n}.$$

Use the logarithm real on the positive real axis, continued to $\operatorname{Re} z > 0$. Put $p_{4,n} = 1$, $p_{3,n} = -4b_n$, $p_{2,n} = q_n$, $p_{1,n} = 4a_n$.

Corollary 3.3 (Normalized primitive). *On $\operatorname{Re} z > 0$, the function*

$$\begin{aligned} \mathcal{P}_4(z) = & -\frac{1}{3z^3} - \frac{2\gamma}{z^2} - \frac{6\gamma^2 - 4\zeta(2)}{z} + 4(\gamma^3 - 3\gamma\zeta(2) + \zeta(3)) \log z + \kappa_4 z \\ & + \sum_{n=1}^{\infty} \sum_{j=1}^4 p_{j,n} F_j(n, z) \end{aligned} \quad (3.23)$$

satisfies $\mathcal{P}'_4(z) = \psi(z)^4$. Its expansion at the singular origin has constant term zero. Consequently, for every real $b > 0$,

$$\operatorname{FP} \int_0^b \psi(x)^4 dx = \mathcal{P}_4(b), \quad \mathcal{P}_4(1) = Q_4,$$

and, for $0 < a < b$, the ordinary integral is $\int_a^b \psi(x)^4 dx = \mathcal{P}_4(b) - \mathcal{P}_4(a)$.

Proof. Each F_j is the primitive vanishing at zero of $(n+z)^{-j} - n^{-j}$. Uniformly on a fixed compact set, $F_j(n, z) = O(n^{-j-1})$. The coefficient growth established in Lemma 3.2 therefore proves normal convergence of the series and its differentiated series. Differentiation gives (3.9). The same estimates give uniform convergence on each disk $|z| \leq r < 1$, including the origin, where every summand vanishes. Thus the displayed singular terms fix the constant coefficient to be zero. $\square$

Corollary 3.4 (Quartic polygamma hierarchy). *For every integer $r \geq 1$, away from the poles of ψ ,*

$$\begin{aligned} & \sum_{r_1+r_2+r_3+r_4=r} \frac{r!}{r_1!r_2!r_3!r_4!} \prod_{j=1}^4 \psi^{(r_j)}(z) \\ &= (-1)^r r! \sum_{n=0}^{\infty} \left\{ \frac{\binom{r+3}{3}}{(n+z)^{r+4}} - \frac{4b_n \binom{r+2}{2}}{(n+z)^{r+3}} + \frac{(r+1)q_n}{(n+z)^{r+2}} + \frac{4a_n}{(n+z)^{r+1}} \right\}. \end{aligned} \quad (3.24)$$

The series converges normally. At $z = 1$ its value is a finite polynomial in $\gamma, \zeta(2), \dots, \zeta(r+1)$.

Proof. Differentiate (3.9) r times. Its constant subtractions disappear, and the resulting terms are bounded by a constant times $n^{-r-1}(1 + \log^3 n)$ on each compact set avoiding the poles. The product rule gives the left side. The last assertion follows from $\psi(1) = -\gamma$ and $\psi^{(k)}(1) = (-1)^{k+1} k! \zeta(k+1)$ for $k \geq 1$. $\square$

3.5 Independent verification and a reproducible tail formula

The supplied program checks 44 exact algebraic identities, including all principal coefficients, the residue differences, the finite summation identity for $1 \leq N \leq 25$, the endpoint-constant cancellation, Laurent coefficient extraction, and the primitive derivatives. Its separate numerical calculations use 100 decimal digits and give

$$Q_4 = 13.510120376015827411606240712830448441932798533301918600986096436 \dots$$

One evaluator uses the arithmetic series with a Binet tail. The other integrates the convergent Taylor series of $(x\psi(x))^4$ near zero and uses ordinary quadrature away from zero; it does not use the harmonic bridge or the Mittag-Leffler expansion. Two choices of the matching point, $1/8$ and $1/16$, agree to the displayed precision. The finest arithmetic result differs from these evaluations by less than $1.40 \cdot 10^{-83}$.

Here is the explicit analytic tail control used in the arithmetic calculation. For $K \geq 0$ set

$$a_K(x) = -\frac{1}{2x} - \sum_{k=1}^K \frac{B_{2k}}{2k x^{2k}}, \quad p_K(x) = \frac{1}{x} + \frac{1}{2x^2} + \sum_{k=1}^K \frac{B_{2k}}{x^{2k+1}}.$$

Binet's integral [8, Equation 5.9.15] and the finite geometric expansion give, for $x > 0$,

$$|\psi(x) - \log x - a_K(x)| \leq E_K x^{-2K-2}, \quad |\psi'(x) - p_K(x)| \leq G_K x^{-2K-3},$$

where $G_K = |B_{2K+2}|$ and $E_K = G_K/(2K+2)$. For the differentiated bound, expand $(1+y)^{-2}$ under the differentiated Binet integral; its Taylor remainder after K terms has magnitude at most $(K+1)y^K$ for $y \geq 0$.

For $N \geq 2$ put $U_{N,K} = 1/(2N) + \sum_{k=1}^K |B_{2k}|/(2kN^{2k})$. Replace $\psi(n)^2 + \psi'(n) - \log^2 n$ in the tail $n \geq N$ by

$$2 \log n a_K(n) + a_K(n)^2 + p_K(n).$$

This finite polynomial in n^{-1} and $\log n$ sums exactly by the first two spectral derivatives of $\zeta(s, N)$. The error in the series itself is bounded by

$$\begin{aligned} & 2E_K \zeta''(2K+3, N) - 2U_{N,K} E_K \zeta'(2K+3, N) \\ & - E_K^2 \zeta'(4K+5, N) - G_K \zeta'(2K+4, N). \end{aligned} \quad (3.25)$$

To obtain the error in Q_4 , multiply this bound by 12. Indeed, if $R = \psi - \log - a_K$, then

$$|\psi^2 + \psi' - (\log + a_K)^2 - p_K| \leq 2(\log n + U_{N,K}) E_K n^{-2K-2} + E_K^2 n^{-4K-4} + G_K n^{-2K-3},$$

and summing after multiplication by $\log n/n$ gives (3.25). With $N = 64$, $K = 34$, the bound for Q_4 is below $1.88 \cdot 10^{-83}$. The implementation uses ordinary mpmath arithmetic, so this analytic truncation bound does not include an interval enclosure of rounding error; the numerical evidence is reported with that qualification.

Scope of the advance. The Mittag-Leffler method, harmonic Laurent coefficients, the Mellin representation, and the Euler sums are classical. The incoming cubic report supplies the immediate proof strategy and explicitly asks for the quartic reduction. The new result relative to the inspected corpus is (3.3), including its exact finite correction, convergent arithmetic form and normalized primitive. The claim is a solution of that stated quartic reduction problem, not a proof that θ is irreducible or a claim of global priority for every equivalent formula.

Further questions. The contour argument extends to each fixed power ψ^k , but the finite arithmetic reduction has only been carried out here through $k = 4$. The next case asks which strict harmonic Laurent coefficients occur in $\text{FP} \int_0^1 \psi(x)^5 dx$ and whether some canonical polynomial in ψ cancels all lower-depth coefficients. A second question is to combine the new quartic bridge with the cubic logarithmic sine moment supplied by the existing beta generator and determine the smallest remaining spectral coordinate space. A third is to obtain a common-Hurwitz-shift version with all boundary terms fixed; simply replacing γ_j by $\gamma_j(a)$ in (3.3) is not justified by this proof.

4 Every centered moment of two even harmonic tails

The latest harmonic-parity report asks for exact centered moments of $(\zeta(2i) - H_n^{(2i)})(\zeta(2j) - H_n^{(2j)})$, beginning with $(i, j) = (1, 2)$. Its all-degree evaluation covers only the trigamma square. The following results answer the question for every pair of even orders and every nonpositive even spectral argument. They also give a convergent generating function in classical polylogarithms. Products of four or more tails, and products with additional powers of H_n , are outside the theorem.

4.1 Definitions, continuation, and an ordinary convergent sum

Throughout this section, $p, q \geq 2$ are even integers, $W = p + q$, and

$$A_p(n) = \zeta(p) - H_n^{(p)} = \zeta(p, n+1) = \frac{\psi^{(p-1)}(n+1)}{(p-1)!}, \quad x_n = n + \frac{1}{2}.$$

The last equality uses that p is even. Define

$$D_{p,q}(s) = \sum_{n=0}^{\infty} \frac{A_p(n)A_q(n)}{x_n^s}, \quad \Re s > 3 - p - q. \quad (4.1)$$

The defining series converges normally in this half-plane. A value outside it always refers to meromorphic continuation unless it is written as an explicitly subtracted convergent sum.

Set

$$c_{p,j} = \frac{1}{p-1} \binom{p+2j-2}{p-2} B_{2j}(1/2), \quad k_{p,q,\ell} = \sum_{j=0}^{\ell} c_{p,j} c_{q,\ell-j}. \quad (4.2)$$

Thus $c_{p,0} = 1/(p-1)$. The standard Hurwitz-zeta expansion gives

$$A_p(n) \sim \sum_{j \geq 0} c_{p,j} x_n^{1-p-2j}, \quad A_p(n)A_q(n) \sim \sum_{\ell \geq 0} k_{p,q,\ell} x_n^{2-W-2\ell}. \quad (4.3)$$

Every exponent in the product is even. The expansions and their fixed-order remainder estimates follow either from the classical Hurwitz formulas or directly from the Laplace integral used below [8].

Proposition 4.1 (Continuation and ordinary summation). *The function $D_{p,q}$ continues meromorphically to the plane. Its only possible poles are simple and lie at*

$$3 - p - q, 1 - p - q, -1 - p - q, \dots$$

In particular it is regular at every $s = -2r$, $r \geq 0$. Let

$$K_r = \max\{0, r - W/2 + 2\}.$$

Then

$$D_{p,q}(-2r) = \sum_{n=0}^{\infty} \left\{ x_n^{2r} A_p(n)A_q(n) - \sum_{\ell=0}^{K_r-1} k_{p,q,\ell} x_n^{2r+2-W-2\ell} \right\}. \quad (4.4)$$

An empty inner sum is zero. Every bracket is $O(n^{-2})$ or smaller, so the right side is an ordinary absolutely convergent sum.

Proof. Subtract any fixed number of terms from the product expansion in (4.3). The remainder series extends to a larger left half-plane; the removed terms sum to $k_{p,q,\ell} \zeta(s + W - 2 + 2\ell, 1/2)$. Increasing the number of subtractions proves the continuation and pole list. At $s = -2r$, the nondecaying terms are exactly those with $0 \leq \ell < K_r$. Their continued sums vanish, because $\zeta(-2m, 1/2) = 0$ for every integer $m \geq 0$. The remaining sum converges absolutely and gives (4.4). $\square$

The half-shift is essential to this argument: it makes both the asymptotic powers and the polynomial subtraction compatible with the trivial zeros of the same Hurwitz zeta function.

4.2 A finite formula at every moment order

Our double-zeta convention is

$$\zeta(a, b) = \sum_{\ell > k \geq 1} \ell^{-a} k^{-b}, \quad a \geq 2, \quad b \geq 1.$$

Only convergent double zeta values occur in the formulas below. For a positive odd integer d , define the ordered contribution

$$U_{p,q}(d) = \begin{cases} \zeta(p, q-d) + \frac{1}{2}\zeta(W-d), & d < q, \\ \sum_{h=1}^{\min\{(d-q+1)/2, p/2-1\}} \frac{\binom{d-q+1}{2h}}{d-q+1} B_{d-q+1-2h} \zeta(p-2h), & d > q. \end{cases} \quad (4.5)$$

The two cases exhaust the possibilities since q is even. The upper limit in the second line is an integer; the sum may be empty. Define also the rational contribution

$$R_{p,q}(d) = \begin{cases} 0, & d < W-1, \\ -\frac{B_{d-W+1}}{2} \left\{ \frac{1}{p} \binom{d-q}{p-1} + \frac{1}{q} \binom{d-p}{q-1} \right\}, & d \geq W-1, \end{cases} \quad (4.6)$$

and

$$C_{p,q}(d) = U_{p,q}(d) + U_{q,p}(d) + R_{p,q}(d). \quad (4.7)$$

These expressions use only finite sums of explicitly specified Bernoulli numbers and ordinary or double zeta values.

Theorem 4.2 (All even orders and all centered moments). *For even $p, q \geq 2$ and every integer $r \geq 0$,*

$$D_{p,q}(-2r) = \sum_{j=0}^r \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_{p,q}(2j+1). \quad (4.8)$$

In particular this formula evaluates the ordinary sums in (4.4).

Proof. Write CT_N for the coefficient of $N^0(\log N)^0$ in an asymptotic expansion at positive infinity. This coefficient will only be taken after exact finite summation. Let

$$S_r(k) = \sum_{n=0}^{k-1} (n+1/2)^{2r} = \frac{B_{2r+1}(k+1/2)}{2r+1} = \sum_{j=0}^r \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) k^{2j+1}. \quad (4.9)$$

This is an odd polynomial in k , with zero constant coefficient. Consequently every polynomial partial sum of a subtracted even power in (4.4) has zero constant coefficient in N . It follows that

$$D_{p,q}(-2r) = \text{CT}_N \sum_{n=0}^{N-1} x_n^{2r} A_p(n) A_q(n). \quad (4.10)$$

The identity $A_p(k-1) = A_p(k) + k^{-p}$ yields finite summation by parts:

$$\begin{aligned} \sum_{n=0}^{N-1} x_n^{2r} A_p(n) A_q(n) &= S_r(N) A_p(N) A_q(N) \\ &\quad + \sum_{k=1}^N S_r(k) \left\{ \frac{A_p(k)}{k^q} + \frac{A_q(k)}{k^p} + \frac{1}{k^{p+q}} \right\}. \end{aligned} \quad (4.11)$$

For a positive odd d , the coefficient block to evaluate is

$$E_{p,q,d}(N) = N^d A_p(N) A_q(N) + \sum_{k=1}^N k^{d-q} A_p(k) + \sum_{k=1}^N k^{d-p} A_q(k) + \sum_{k=1}^N k^{d-W}. \quad (4.12)$$

We prove that its constant coefficient is $C_{p,q}(d)$.

First suppose $d < q$. The corresponding tail sum converges, and its limit is exactly

$$\sum_{k \geq 1} k^{d-q} A_p(k) = \zeta(p, q-d). \quad (4.13)$$

No continued or divergent double sum is used. If $m = d - q \geq 1$, then m is odd. Put

$$F_m(N) = \sum_{k=1}^N k^m = \frac{B_{m+1}(N+1) - B_{m+1}}{m+1}.$$

Interchanging a finite sum gives the exact identity

$$\sum_{k=1}^N k^m A_p(k) = A_p(N) F_m(N) + \sum_{\ell=1}^N \frac{F_m(\ell-1)}{\ell^p}. \quad (4.14)$$

Since

$$F_m(\ell-1) = \frac{1}{m+1} \sum_{a=1}^{m+1} \binom{m+1}{a} B_{m+1-a} \ell^a,$$

the constant term of the last sum is obtained directly from finite power sums. Exponents $a-p \geq 0$ give zero constant; exponent -1 gives γ ; smaller exponents give $\zeta(p-a)$. Because m is odd, the nonzero Bernoulli coefficients have a even, except for the half-leading term $a = m$, whose coefficient is $-1/2$. Thus the even values contribute

$$\sum_{h=1}^{\min\{(m+1)/2, p/2-1\}} \frac{\binom{m+1}{2h}}{m+1} B_{m+1-2h} \zeta(p-2h). \quad (4.15)$$

If $d < W-1$, the half-leading term contributes $-\zeta(W-d)/2$. At $d = W-1$, it instead contributes $-\gamma/2$. If $d > W-1$, it contributes zero.

The final power sum in (4.12) contributes $\zeta(W-d)$ for $d < W-1$, γ for $d = W-1$, and zero for $d > W-1$. Combining it with the two half-leading terms has a useful consequence. Each ordered block with $d < q$ retains one half of $\zeta(W-d)$, while every block with $d > q$ retains none. At $d = W-1$, the two terms $-\gamma/2$ cancel the final $+\gamma$ exactly. Equations (4.13) and (4.15) therefore give precisely $U_{p,q}(d) + U_{q,p}(d)$.

It remains to compute the rational boundary contributions. The uncentered expansion is

$$A_p(N) \sim \frac{N^{1-p}}{p-1} - \frac{1}{2} N^{-p} + \sum_{j \geq 1} \frac{1}{p-1} \binom{p+2j-2}{p-2} B_{2j} N^{1-p-2j}. \quad (4.16)$$

For any odd integer k , define the inverse-power coefficient by

$$u_p(k) = \begin{cases} 0, & k < p-1, \\ \frac{1}{p-1} \binom{k-1}{p-2} B_{k-p+1}, & k \geq p-1. \end{cases}$$

The only nonzero even inverse-power coefficient in (4.16) is $-1/2$, at exponent p . In the Faulhaber polynomial $F_m(N)$, all powers are even except for its half-leading term $N^m/2$. Pairing the possible exponents therefore gives, for odd $m \geq 1$,

$$\text{CT}_N \{A_p(N) F_m(N)\} = \frac{p-m}{2p} u_p(m). \quad (4.17)$$

Indeed the two potential contributions are $u_p(m)/2$ and $-\binom{m+1}{p}B_{m+1-p}/(2(m+1))$; the latter equals $-mu_p(m)/(2p)$. If $m < p - 1$, both are zero.

The odd coefficient of the product in the first term of (4.12) is

$$\text{CT}_N\{N^d A_p(N) A_q(N)\} = -\frac{1}{2}\{u_p(d-q) + u_q(d-p)\},$$

where a term below its support is zero. Adding this to the two contributions (4.17) gives

$$-\frac{d-q}{2p}u_p(d-q) - \frac{d-p}{2q}u_q(d-p).$$

It vanishes for $d < W - 1$. Otherwise, the binomial identity

$$\frac{d-q}{p-1} \binom{d-q-1}{p-2} = \binom{d-q}{p-1}$$

reduces it to $R_{p,q}(d)$ in (4.6). We have proved $\text{CT}_N E_{p,q,d}(N) = C_{p,q}(d)$. Combining (4.9), (4.10), and (4.11) proves the theorem. $\square$

4.3 Reduction to a fixed set of ordinary zeta coordinates

There is no unresolved depth-two constant in the preceding theorem. Every double zeta appearing in (4.5) has even outer order, odd inner order, and odd total weight. The classical odd-weight reduction proved by Borwein, Borwein, and Girgensohn [9] gives the following explicit elimination. For even $a \geq 2$, odd $b \geq 3$, and $w = a + b$,

$$\begin{aligned} \zeta(a, b) &= \frac{1}{2} \left\{ \binom{w}{a} - 1 \right\} \zeta(w) + \zeta(a) \zeta(b) \\ &\quad - \sum_{k=1}^{(w-3)/2} \left\{ \binom{w-2k-1}{b-1} + \binom{w-2k-1}{a-1} \right\} \zeta(2k) \zeta(w-2k). \end{aligned} \quad (4.18)$$

For the boundary case $b = 1$, Euler's formula is

$$\zeta(a, 1) = \frac{a}{2} \zeta(a+1) - \sum_{k=1}^{a/2-1} \zeta(2k) \zeta(a+1-2k). \quad (4.19)$$

All zeta arguments on the right are at least two. These are classical inputs, not novelty claims of this article.

Corollary 4.3 (A moment-independent coordinate space). *Let $M = \max\{p, q\}$. Every value $D_{p,q}(-2r)$, for arbitrary $r \geq 0$, lies in the same rational span*

$$\text{span}_{\mathbb{Q}}(\{1\} \cup \{\zeta(2h) : 1 \leq h \leq M/2 - 1\} \cup \{C_{p,q}(d) : 1 \leq d < M, d \text{ odd}\}). \quad (4.20)$$

This is a spanning set of M specified numbers, each explicitly reducible to ordinary zeta values and their pairwise products. It is not a claim that those numbers are linearly independent.

Proof. For every odd $d > M$, both ordered blocks in (4.5) contain only even zeta values of argument at most $M - 2$, and the remaining block is rational. The odd integers $d < M$ give the finite list in (4.20). The moment formula proves the inclusion, while (4.18) and (4.19) give the stated ordinary-zeta reduction. $\square$

This finiteness is uniform in the moment order. Increasing r changes only rational Bernoulli coefficients; it introduces no new arithmetic constants.

4.4 The first mixed pair: a complete $(2, 4)$ evaluation

For $(p, q) = (2, 4)$, write $D_r = D_{2,4}(-2r)$. In this case the entire block sequence has the short form

$$\begin{aligned} C_{2,4}(1) &= \frac{15}{2}\zeta(5) - 3\zeta(2)\zeta(3), \\ C_{2,4}(3) &= \frac{3}{2}\zeta(3) + \frac{1}{2}\zeta(2), \\ C_{2,4}(2j+1) &= \frac{2j-1}{2}B_{2j-2}\zeta(2) - \frac{(2j-3)(2j^2-3j+7)}{24}B_{2j-4}, \quad j \geq 2. \end{aligned} \quad (4.21)$$

For example, the first line follows from $\zeta(2, 3) = 9\zeta(5)/2 - 2\zeta(2)\zeta(3)$ and $\zeta(4, 1) = 2\zeta(5) - \zeta(2)\zeta(3)$. Thus the arithmetic space in this case can be written especially economically as

$$\text{span}_{\mathbb{Q}}\{1, \zeta(2), \zeta(3), 5\zeta(5) - 2\zeta(2)\zeta(3)\}. \quad (4.22)$$

The first four values are

$$\begin{aligned} D_0 &= \frac{15}{2}\zeta(5) - 3\zeta(2)\zeta(3), \\ D_1 &= -\frac{5}{8}\zeta(5) + \frac{1}{4}\zeta(2)\zeta(3) + \frac{1}{2}\zeta(3) + \frac{1}{6}\zeta(2), \\ D_2 &= \frac{7}{32}\zeta(5) - \frac{7}{80}\zeta(2)\zeta(3) - \frac{1}{4}\zeta(3) - \frac{1}{30}\zeta(2) - \frac{3}{40}, \\ D_3 &= -\frac{155}{896}\zeta(5) + \frac{31}{448}\zeta(2)\zeta(3) + \frac{7}{32}\zeta(3) - \frac{1}{672}\zeta(2) + \frac{31}{672}. \end{aligned} \quad (4.23)$$

The first two sums converge without subtraction. The next two yield the polygamma identities

$$\begin{aligned} \sum_{n=0}^{\infty} \left\{ \frac{x_n^4}{6} \psi'(n+1) \psi'''(n+1) - \frac{1}{3} \right\} &= D_2, \\ \sum_{n=0}^{\infty} \left\{ \frac{x_n^6}{6} \psi'(n+1) \psi'''(n+1) - \frac{x_n^2}{3} + \frac{7}{36} \right\} &= D_3. \end{aligned} \quad (4.24)$$

The signs and normalizations of both subtractions follow directly from $k_{2,4,0} = 1/3$ and $k_{2,4,1} = -7/36$.

There is also an exact finite-generation statement. The inverse Bernoulli transform is

$$C_{p,q}(2j+1) = \sum_{r=0}^j \binom{2j+1}{2r} 2^{-2(j-r)} D_{p,q}(-2r). \quad (4.25)$$

It follows by multiplying the generating function in the next section by $2 \sinh(t/2)/t$, or directly by the Bernoulli generating identity. Put $C_j = C_{2,4}(2j+1)$. Then

$$C_2 = \frac{1}{4}\zeta(2) - \frac{3}{8}, \quad C_3 = -\frac{1}{12}\zeta(2) - \frac{1}{3},$$

so

$$1 = -\frac{8}{11}(C_2 + 3C_3), \quad \zeta(2) = \frac{32C_2 - 36C_3}{11}, \quad \zeta(3) = \frac{2}{3}C_1 - \frac{32}{33}C_2 + \frac{12}{11}C_3.$$

Consequently *every* D_r is an explicit rational linear combination of D_0, D_1, D_2, D_3 . This is an exact generation statement, independent of any conjectured arithmetic independence. Two useful eliminations are

$$\begin{aligned} 7D_0 + 412D_1 + 2000D_2 + 1344D_3 &= -88, \\ 23D_0 + 524D_1 - 2480D_2 - 4032D_3 &= 176\zeta(2). \end{aligned} \quad (4.26)$$

For instance the first is the single absolutely convergent identity

$$\sum_{n=0}^{\infty} \left\{ (1344x_n^6 + 2000x_n^4 + 412x_n^2 + 7) \frac{\psi'(n+1) \psi'''(n+1)}{6} - 448x_n^2 - \frac{1216}{3} \right\} = -88. \quad (4.27)$$

5 A classical-polylogarithm generator for every pair

The finite formula of Theorem 4.2 can be summed at every moment order. This yields an analytic identity, as opposed to a merely formal generating series.

For integers $a \geq 2$ and $b \geq 1$, define

$$I_{a,b}(t) = \int_0^t u^a(t-u)^b \coth(u/2) du. \quad (5.1)$$

Inside $|t| < 2\pi$, use the straight segment from zero to t . The integrand has a removable zero-endpoint singularity, and the integral is unambiguous there. Define the finite polynomial

$$P_{p,q}(t) = \sum_{\substack{1 \leq d < q \\ d \text{ odd}}} \left\{ \zeta(p, q-d) + \frac{1}{2} \zeta(W-d) \right\} \frac{t^{d-1}}{d!}. \quad (5.2)$$

Every double zeta in this polynomial is eliminated by (4.18) or (4.19).

Theorem 5.1 (All-pair analytic moment generator). *For even $p, q \geq 2$, the series*

$$G_{p,q}(t) = \sum_{r=0}^{\infty} D_{p,q}(-2r) \frac{t^{2r}}{(2r)!}$$

converges in $|t| < 2\pi$. In that disk it is the even holomorphic function

$$\boxed{G_{p,q}(t) = \frac{t}{2 \sinh(t/2)} \{F_{p,q}(t) + F_{q,p}(t)\}}, \quad (5.3)$$

where all singularities at zero are removable and

$$\begin{aligned} F_{p,q}(t) &= P_{p,q}(t) \\ &+ \sum_{h=1}^{p/2-1} \frac{\zeta(p-2h)}{2t(2h)!(q-1)!} I_{2h,q-1}(t) - \frac{I_{p,q-1}(t)}{4tp!(q-1)!}. \end{aligned} \quad (5.4)$$

For real $0 < t < 2\pi$, the expression in braces has a finite form using only rational functions of t , ordinary zeta values, and $\text{Li}_k(e^{-t})$ for $2 \leq k \leq p+q$. The full generator includes the displayed elementary prefactor $t/(2 \sinh(t/2))$. The integral formula fixes its holomorphic continuation at zero.

Proof. The Bernoulli convolution in (4.8) is equivalent formally to

$$G_{p,q}(t) = \frac{t}{2 \sinh(t/2)} \sum_{\substack{d \geq 1 \\ d \text{ odd}}} C_{p,q}(d) \frac{t^{d-1}}{d!}. \quad (5.5)$$

We show that the last series converges in the stated disk and equals $F_{p,q} + F_{q,p}$.

Write

$$E(u) = \frac{u}{2} \coth(u/2) = \sum_{k \geq 0} \frac{B_{2k}}{(2k)!} u^{2k}, \quad |u| < 2\pi.$$

The terms with $d < q$ in the ordered block are exactly $P_{p,q}$. For each fixed $1 \leq h \leq p/2 - 1$, put $d = q + 2h - 1 + 2k$. Its remaining Bernoulli contribution is

$$\frac{\zeta(p-2h)}{(2h)!} \sum_{k \geq 0} \frac{(2h+2k-1)!}{(q+2h+2k-1)!} \frac{B_{2k}}{(2k)!} t^{q+2h+2k-2}.$$

Euler's beta integral transforms this normally convergent series into

$$\frac{\zeta(p-2h)}{t(2h)!(q-1)!} \int_0^t u^{2h-1}(t-u)^{q-1} E(u) \, du = \frac{\zeta(p-2h)I_{2h,q-1}(t)}{2t(2h)!(q-1)!}.$$

For the first binomial term in $R_{p,q}(d)$, set $d = W - 1 + 2k$. Its series is

$$-\frac{1}{2p!} \sum_{k \geq 0} \frac{(p+2k-1)!}{(W+2k-1)!} \frac{B_{2k}}{(2k)!} t^{W+2k-2},$$

which the same beta integral turns into $-I_{p,q-1}(t)/(4tp!(q-1)!)$. The second term is the same expression with p, q exchanged. This proves (5.4) and (5.3).

All power-series integrations above are normal on compact subsets of $|t| < 2\pi$. The prefactor $t/(2 \sinh(t/2))$ is holomorphic there, including at zero. The resulting function is even: this follows from its Taylor construction, or by substituting $u = tv$ in each integral. Thus (5.5) is the Taylor series of an even holomorphic function, proving the claimed convergence.

Finally, write $\coth(u/2) = 1 + 2/(e^u - 1)$ and expand the polynomial $(t - u)^b$. For $t > 0$, elementary integration gives

$$\begin{aligned} I_{a,b}(t) &= \frac{a!b!}{(a+b+1)!} t^{a+b+1} \\ &\quad + 2 \sum_{\ell=0}^b (-1)^\ell \binom{b}{\ell} t^{b-\ell} J_{a+\ell}(t), \\ J_m(t) &= m! \left\{ \zeta(m+1) - \sum_{j=0}^m \frac{t^j}{j!} \text{Li}_{m+1-j}(e^{-t}) \right\}. \end{aligned} \quad (5.6)$$

One proof expands $(e^u - 1)^{-1}$ into its positive exponential series and integrates each term. Since $m \geq 2$, the integrals are absolutely convergent. The coefficient of $\text{Li}_1(e^{-t})$ in the first two lines is proportional to $\sum_{\ell=0}^b (-1)^\ell \binom{b}{\ell} = 0$. The largest remaining polylogarithm order is $a + b + 1 \leq p + q$. This proves the last assertion. $\square$

5.1 An explicit Li_2 – Li_6 identity

For $(p, q) = (2, 4)$, the general generator becomes

$$\begin{aligned} G_{2,4}(t) &= \frac{t}{2 \sinh(t/2)} \left\{ \frac{15}{2} \zeta(5) - 3\zeta(2)\zeta(3) + \frac{t^2}{4} \zeta(3) + \frac{\zeta(2)}{4t} I_{2,1}(t) \right. \\ &\quad \left. - \frac{I_{4,1}(t)}{96t} - \frac{I_{2,3}(t)}{48t} \right\}. \end{aligned} \quad (5.7)$$

Put $L_k(t) = \text{Li}_k(e^{-t})$. An entirely explicit version, valid for real $0 < t < 2\pi$, is

$$G_{2,4}(t) = \frac{t}{2 \sinh(t/2)} \mathcal{L}_{2,4}(t), \quad (5.8)$$

where

$$\begin{aligned} \mathcal{L}_{2,4}(t) &= 4\zeta(5) - 2\zeta(2)\zeta(3) + \frac{t^2}{6} \zeta(3) + \frac{3t}{4} \zeta(4) + \frac{t^3}{48} \zeta(2) - \frac{t^5}{1440} \\ &\quad + \left(\frac{t}{2} \zeta(2) - \frac{t^3}{48} \right) L_2(t) + \left(2\zeta(2) - \frac{t^2}{6} \right) L_3(t) \\ &\quad - tL_4(t) - 4L_5(t) + \frac{3\zeta(2)}{t} \{L_4(t) - \zeta(4)\} + \frac{15}{2t} \{\zeta(6) - L_6(t)\}. \end{aligned} \quad (5.9)$$

Although individual polylogarithms have logarithmic expansions at zero, their specified combination is even and holomorphic there. Equation (5.7), rather than a separate choice of logarithm for each term, provides the continuation. The coefficient of $t^{2r}/(2r)!$ is exactly the all-order moment in (4.8) and (4.21).

5.2 Scope and next exact questions

The two-tail problem is closed by a finite Bernoulli formula, ordinary-zeta elimination, and a convergent classical-polylogarithm generator. The next questions are structurally different.

1. For four even tails, determine the smallest useful multiple-zeta coordinate space uniform in the centered moment order. Finite summation by parts still lowers the number of tails, but higher-depth constants can enter. A two-tail calculation does not prove their elimination.
2. Determine exact first spectral derivatives at the regular points $s = -2r$. These contain logarithmically weighted remainder sums in addition to local Bernoulli data. The value theorem alone does not reduce those global contributions.
3. Introduce rational inner shifts while retaining a matched center for the outer weight. Identify the resulting finite colored Euler-sum coordinates and the precise analogues of the classical-polylogarithm generator.
4. For fixed p, q , find optimal finite generating sets of the actual moment sequence. Corollary 4.3 provides at most $\max\{p, q\}$ explicit coordinates, and $(2, 4)$ has an exact four-moment generator. Arithmetic independence of a proposed minimal set must be distinguished from the rank of its rational coefficient matrix.

6 Complex powers of a cotangent resolvent

The rational resolvent theorem in the incoming harmonic-parity report leaves open its extension to complex powers (Question R4). The complex-power sine transform in the higher-reflection report treats a different, reflection-symmetric weight. We now give a one-sided complex-power transform at every nonreal resolvent parameter, an ordinary Euler integral involving a single classical polylogarithm, and a completed germ at every integral power. The latter includes a removable but nonzero endpoint correction when the integral power is negative. Classical Hurwitz Fourier analysis and beta integrals supply the analytic tools [8, 10]; the formulas and their normalizations below are proved directly.

6.1 A branch fixed by the interval

For $z \in \mathbb{C} \setminus \mathbb{R}$, put

$$\epsilon = \operatorname{sgn} \Im z, \quad a = z + \epsilon\pi i, \quad q = \frac{z - \epsilon\pi i}{a}, \quad \sigma = -2\epsilon\pi i, \quad C = -\frac{1}{a} = \frac{1-q}{\sigma}. \quad (6.1)$$

Thus $|q| < 1$. Throughout,

$$\operatorname{Log} \sigma = \log(2\pi) - \epsilon\pi i/2, \quad \operatorname{Log} C = \operatorname{Log}(1-q) - \operatorname{Log} \sigma.$$

Both $\operatorname{Log} C$ and $\operatorname{Log}(1-q)$ in these equations are principal logarithms. Write

$$K(x) = \pi \cot(\pi x), \quad R_z(x) = \frac{1}{K(x) - z}, \quad w = e^{2\epsilon\pi i x}. \quad (6.2)$$

Direct algebra gives

$$R_z(x) = C \frac{1-w}{1-qw}. \quad (6.3)$$

The continuous logarithm of $R_z(x)$ on $0 < x < 1$ is fixed by $\text{Log } R_z(x) - \log x \rightarrow 0$ at zero. Its imaginary part lies between 0 and $\epsilon\pi$, and it approaches $\epsilon\pi$ at one. In particular,

$$R_z(x)^\lambda = \exp(\lambda \text{Log } R_z(x)), \quad R_z(1-y)^\lambda = e^{\epsilon\pi i \lambda} y^\lambda V_z(-y)^\lambda, \quad V_z(x) = \frac{R_z(x)}{x}. \quad (6.4)$$

Here V_z is analytic near zero and $V_z(0) = 1$. Powers of V_z use the analytic logarithm vanishing at zero. These prescriptions also specify all logarithmic derivatives in λ .

6.2 A beta mixture of ordinary polylogarithms

Define coefficients and their Dirichlet transform by

$$\left(\frac{1-w}{1-qw} \right)^\lambda = \sum_{n \geq 0} d_n(\lambda, q) w^n, \quad d_0 = 1, \quad (6.5)$$

$$d_n(\lambda, q) = \sum_{j=0}^n \frac{(-\lambda)_j (\lambda)_{n-j}}{j!(n-j)!} q^{n-j}, \quad (6.6)$$

$$\mathcal{L}_\lambda(\nu, q) = \sum_{n \geq 1} \frac{d_n(\lambda, q)}{n^\nu}. \quad (6.7)$$

The last series is initially normally convergent for $\Re(\nu + \lambda) > 0$, on compact parameter sets away from the boundary of $|q| < 1$. At integral λ it can have a larger convergence domain; our continuation will not rely on such exceptional improvements. The notation in (6.7) is a transform of the explicit coefficients (6.6), not a claim that an unevaluated integral has been reduced to familiar constants.

Theorem 6.1 (The Euler–polylogarithm formula). *For $-1 < \Re \lambda < 1$, $|q| < 1$, and initially $\Re \nu > 1$,*

$$\mathcal{L}_\lambda(\nu, q) = -\frac{\sin \pi \lambda}{\pi} (1-q) \int_0^1 y^{-\lambda} (1-y)^\lambda \frac{\text{Li}_\nu(q + (1-q)y)}{q + (1-q)y} dy. \quad (6.8)$$

The quotient $\text{Li}_\nu(t)/t$ has value 1 at $t = 0$. Thus the formula remains valid when the segment from q to 1 passes through zero. Within the same λ strip the formula also holds on $\Re(\nu + \lambda) > 0$ by analytic continuation of the convergent integrals. All fixed parameter derivatives commute with these integrals on compact subsets of their convergence domains.

Proof. We first prove the coefficient identity

$$d_n(\lambda, q) = -\frac{\sin \pi \lambda}{\pi} (1-q) \int_0^1 y^{-\lambda} (1-y)^\lambda (q + (1-q)y)^{n-1} dy, \quad n \geq 1. \quad (6.9)$$

For $0 < q < 1$, deform the Cauchy coefficient contour for $((1-w)/(1-qw))^\lambda$ onto $1 < w < 1/q$. The difference of its upper and lower boundary values is $-2i \sin(\pi \lambda) ((w-1)/(1-qw))^\lambda$. The endpoint integrals exist precisely in the stated strip. The integral at infinity vanishes for $n \geq 1$. Substituting $t = 1/w$ and then $t = q + (1-q)y$ proves (6.9). Alternatively, its right side can be expanded as a finite beta sum and checked against (6.6). Both sides are polynomials in q , so the identity holds for all complex q .

For $\Re \nu > 1$ we may sum (6.9) after multiplying by $n^{-\nu}$. The segment lies in the closed unit disk and only its last endpoint can meet the unit circle. The absolutely convergent polylogarithm

series, together with the integrable beta weight, provides domination. This yields (6.8). Near $y = 1$ the possible singular part of $\text{Li}_\nu(q + (1-q)y)$ has order $(1-y)^{\nu-1}$, with the usual logarithmic limits at integral orders [8, Eq. 25.12.12]. The conditions $\Re\lambda > -1$ and $\Re(\nu + \lambda) > 0$ therefore give the asserted extension and the same domination after any fixed number of derivatives. Near $y = 0$, $\Re\lambda < 1$ suffices, including $q = 0$. $\square$

There is a useful independent Mellin representation:

$$\mathcal{L}_\lambda(\nu, q) = \frac{1}{\Gamma(\nu)} \int_0^\infty t^{\nu-1} \left\{ \left(\frac{1-e^{-t}}{1-qe^{-t}} \right)^\lambda - 1 \right\} dt, \quad (6.10)$$

initially for $\Re\nu > \max(0, -\Re\lambda)$. At zero, the expression in braces is t^λ times an analytic function minus 1; at infinity it decreases exponentially. Finite Taylor subtraction in (6.10) proves joint meromorphic continuation in (ν, λ) . Its only possible moving polar hyperplanes are $\nu + \lambda = -j$ for $j \in \mathbb{Z}_{\geq 0}$; some disappear through zeros of $1/\Gamma(\nu)$ or of their coefficients. This supplies a continuation independently of the Euler integral's restricted λ strip.

6.3 The Hurwitz transform and an ordinary log-Gamma integral

Theorem 6.2 (Complex-power Hurwitz transform). *For $p \in \mathbb{Z}_{\geq 0}$, $\Re\lambda > -1$, and $\Re(u + \lambda) > p$, away from the Hurwitz pole when $p = 0$,*

$$\boxed{\int_0^1 \partial_x^p \zeta(1-u, x) R_z(x)^\lambda dx = C^\lambda \Gamma(u) \sigma^{p-u} \mathcal{L}_\lambda(u-p, q).} \quad (6.11)$$

The identity then continues meromorphically in its parameters. Also, for every $\Re\lambda > -1$,

$$\int_0^1 R_z(x)^\lambda dx = C^\lambda. \quad (6.12)$$

Proof. For $\Re u > p + 1$, the Hurwitz Fourier series and its first p argument derivatives are absolutely convergent. The nonzero Fourier pairings are

$$\int_0^1 \partial_x^p \zeta(1-u, x) e^{2\epsilon\pi i n x} dx = \Gamma(u) \sigma^{p-u} n^{p-u}, \quad n \geq 1,$$

and the zero mode vanishes. Multiply by the Abel-regularized expansion (6.5) and then let the Abel radius tend to one. For $\Re\lambda > -1$ its boundary values are uniformly integrable: near either endpoint they are bounded by a constant times $x^{\min(\Re\lambda, 0)}$ or $(1-x)^{\min(\Re\lambda, 0)}$. This proves the formula in an initial open region. The Dirichlet series is normally convergent under $\Re(u + \lambda) > p$. The original integral has the same sufficient local conditions, since its only nonsmooth Hurwitz term at zero is $\prod_{j=1}^p (u-j)x^{u-p-1}$. Holomorphic continuation within that domain proves (6.11). The constant Fourier coefficient of the Abel kernel is C^λ ; the same integrable limit proves (6.12). $\square$

Corollary 6.3 (An ordinary Gamma moment). *For $\Re\lambda > -1$,*

$$\int_0^1 \log \Gamma(x) R_z(x)^\lambda dx = C^\lambda \left\{ \frac{\log(2\pi)}{2} + \frac{(\gamma + \text{Log } \sigma) \mathcal{L}_\lambda(1, q) - \partial_\nu \mathcal{L}_\lambda(1, q)}{\sigma} \right\}. \quad (6.13)$$

Proof. Differentiate (6.11) with $p = 0$ at $u = 1$ and use $\zeta'(0, x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$ and $\Gamma'(1) = -\gamma$ [8]. The logarithmic endpoint singularity is integrable under the stated condition, so this is an ordinary integral and an ordinary derivative. $\square$

Thus, in the strip of Theorem 6.1, the Gamma moment is an explicit combination of two integrals of Li_1 and its first order derivative against an elementary beta weight. At $q = 0$, its first transform is $\mathcal{L}_\lambda(1, 0) = -\psi(1 + \lambda) - \gamma$. Section 7 gives the stronger finite Gamma generating formula for all polynomial moments.

6.4 Completion at every integral power

Let

$$g(u, x) = \zeta(1 - u, x) + \frac{1}{u} = \sum_{m \geq 0} \gamma_m(x) \frac{u^m}{m!}, \quad A_p(u) = \prod_{j=1}^p (u - j), \quad A_0 = 1. \quad (6.14)$$

Define finite local coefficients

$$v_j(\lambda; z) = [x^j] V_z(x)^\lambda, \quad (6.15)$$

$$h_p(u, x) = \partial_x^p \left\{ \zeta(1 - u, 1 + x) + \frac{1}{u} \right\}, \quad (6.16)$$

$$b_j(u, \lambda; z) = [x^j] h_p(u, x) V_z(x)^\lambda. \quad (6.17)$$

The function h_p is holomorphic near $(u, x) = (0, 0)$. Its coefficients are explicit: the coefficient of x^j is

$$\frac{(-1)^{p+j} (1 - u)_{p+j}}{j!} \zeta(1 - u + p + j) \quad (p + j > 0), \quad (6.18)$$

and is $\zeta(1 - u) + 1/u$ when $p = j = 0$. The first coefficients of V_z are

$$V_z(x) = 1 + zx + (z^2 + \pi^2/3)x^2 + (z^3 + 2\pi^2 z/3)x^3 + O(x^4).$$

All fixed Taylor jets of v_j and b_j required for coefficient extraction are therefore finite polynomials in z , the expansion parameters, zeta values or their derivatives, and Stieltjes coefficients. No new global integral enters them.

For an integer L , set $\lambda = L + v$ and

$$\mathcal{T}_p(u, \lambda; z) = C^\lambda \Gamma(u) \sigma^{p-u} \mathcal{L}_\lambda(u - p, q) + \mathbf{1}_{p=0} \frac{C^\lambda}{u}, \quad (6.19)$$

$$\begin{aligned} \mathcal{H}_{p,L}(u, v; z) &= \mathcal{T}_p(u, L + v; z) - \mathbf{1}_{L \leq p} \frac{A_p(u) v_{p-L}(L + v; z)}{u + v} \\ &\quad - \mathbf{1}_{L \leq -1} \frac{\{1 + (-1)^{-L-1} e^{\epsilon \pi i (L+v)}\} b_{-L-1}(u, L + v; z)}{v}. \end{aligned} \quad (6.20)$$

A term with a false indicator is omitted completely. In the last line the quotient is read by its removable analytic extension.

Theorem 6.4 (Joint Stieltjes and power completion). *For each $p \geq 0$, $L \in \mathbb{Z}$, and $z \notin \mathbb{R}$, $\mathcal{H}_{p,L}$ is holomorphic near $(u, v) = (0, 0)$. For all integers $m, k \geq 0$,*

$$\boxed{\text{FP} \int_0^1 \gamma_m^{(p)}(x) R_z(x)^L (\text{Log } R_z(x))^k \, dx = m!k! [u^m v^k] \mathcal{H}_{p,L}(u, v; z).} \quad (6.21)$$

The finite part is taken in the fixed coordinates x and $1 - x$. All coefficient extractions and all z derivatives are locally holomorphic on each open half-plane.

Proof. The exact local splitting is

$$\partial_x^p g(u, x) = A_p(u) x^{u-p-1} + h_p(u, x).$$

At zero, multiplication by $R_z(x)^\lambda = x^\lambda V_z(x)^\lambda$ produces the two families

$$A_p(u)v_j(\lambda; z)x^{u+\lambda-p-1+j}, \quad b_j(u, \lambda; z)x^{\lambda+j}.$$

After setting $\lambda = L + v$, their integration denominators pass through zero only for $j = p - L \geq 0$ and $j = -L - 1 \geq 0$, respectively. They give the second line of (6.20) and the first endpoint's contribution to its third line.

At the other endpoint put $y = 1 - x$. The smooth Hurwitz factor is exactly $h_p(u, -y)$ and (6.4) applies. Its coefficient of y^j is $e^{\epsilon\pi i\lambda}(-1)^j b_j(u, \lambda; z)$. This supplies precisely the remaining term in the braces in (6.20). Their sum vanishes on $v = 0$, since $(-1)^{-L-1}e^{\epsilon\pi iL} = -1$. Nevertheless its quotient by v must be subtracted with the complete numerator.

To make the argument analytic, choose a small endpoint interval and subtract finitely many of the displayed monomials, enough to leave a uniformly integrable remainder throughout a small parameter polydisk. The middle interval is holomorphic. Nonresonant local denominators are bounded away from zero. For a resonant monomial with exponent $\alpha - 1$, the difference between its meromorphic integral and the generating function of its coordinate finite parts is its full amplitude divided by α , because

$$\frac{\delta^\alpha}{\alpha} = \frac{1}{\alpha} + \frac{\delta^\alpha - 1}{\alpha}.$$

The second summand is entire in α and its derivatives give the constants of the corresponding logarithmic integrals. This also shows that the arbitrary splitting point cancels out. Finally, the global Hurwitz pole contributes $-C^\lambda/u$ only for $p = 0$, and (6.19) has already removed it by (6.12). The remaining germ is holomorphic and its coefficients are exactly (6.21). The same normally convergent construction is uniform on compact subsets of either z half-plane. $\square$

Away from integral powers there is no resonant exponent at $u = 0$. Consequently, for $\lambda \notin \mathbb{Z}$, the same local subtraction proof gives the simpler identity

$$\text{FP} \int_0^1 \gamma_m^{(p)}(x) R_z(x)^\lambda (\text{Log } R_z(x))^k dx = m! [u^m] \partial_\lambda^k \mathcal{T}_p(u, \lambda; z). \quad (6.22)$$

Here the finite part means the constant term in the generalized power–logarithm endpoint expansion. For nonintegral λ , none of its subtracted endpoint powers has exponent zero after integration, so differentiation and this finite part commute locally in λ . At integral powers the full completion (6.20), rather than a limit of (6.22), specifies the coordinate finite part.

The correction at negative powers is not zero. Although the last quotient in (6.20) is removable, its value at $v = 0$ is

$$-\epsilon\pi i b_{-L-1}(u, L; z), \quad L \leq -1. \quad (6.23)$$

Removing only pole residues would miss it. It records the different endpoint phases of R_z^λ before λ is specialized to an integer. At a fixed negative integer the integrand itself is a polynomial in $K(x) - z$, so the resulting finite parts must have the corresponding polynomial dependence on z . Formula (6.23) is necessary for that compatibility.

The same issue already appears without a Hurwitz factor. For $M \geq 1$,

$$\text{FP} \int_0^1 (K(x) - z)^M dx = \frac{(-z + i\pi)^M + (-z - i\pi)^M}{2}, \quad (6.24)$$

whereas the continued mean in (6.12) is $C^{-M} = (-z - \epsilon i\pi)^M$. Reflection gives zero for odd cotangent powers, and integration of their derivative recurrence gives $\text{FP} \int_0^1 K^{2j} dx = (-\pi^2)^j$, proving (6.24). The difference is exactly $\epsilon\pi i [x^{M-1}] V_z(x)^{-M}$, as the two-endpoint completion predicts. This is an elementary check at every negative integral power. An independent all- M residue proof is given in Appendix B.3.

Example 6.5 (Two normalization checks). For $L = 0$ and $k = 0$, the theorem gives $\text{FP} \int_0^1 \gamma_m^{(p)}(x) dx = 0$. For $L = -1$, $p = m = k = 0$, it gives

$$\text{FP} \int_0^1 \gamma_0(x)(\pi \cot \pi x - z) dx = -\frac{\pi^2}{2}. \quad (6.25)$$

Indeed $\mathcal{L}_{-1}(u, q) = (1 - q)\zeta(u)$, $v_1(-1; z) = -z$, and $b_0 = \zeta(1 - u) + 1/u$. At $v = 0$, (6.20) reduces, by the zeta functional equation, to

$$2\pi\Gamma(u)(2\pi)^{-u} \sin(\pi u/2)\zeta(u),$$

whose value at zero is $-\pi^2/2$. The same value follows directly from digamma reflection and $\text{FP} \int_0^1 K(x)^2 dx = -\pi^2$. For $L = 1$ and $p = 1$, the moving term is $(u - 1)/(u + v)$; at $z = i\pi$ it recovers

$$\text{FP} \int_0^1 \frac{\psi'(x)}{\pi \cot \pi x - i\pi} dx = 1 - \gamma - \log(2\pi) + i\pi/2,$$

including the inherited resolvent theorem's harmonic correction.

6.5 An explicit logarithmic resolvent identity

At first order in λ there is no need to retain the transform $\mathcal{L}$: differentiating (6.5) at zero gives

$$\partial_\lambda \mathcal{L}_\lambda(u, q)|_{\lambda=0} = \text{Li}_{1+u}(q) - \zeta(1 + u). \quad (6.26)$$

Corollary 6.6 (All Stieltjes indices with one logarithm). *The expression in braces below is holomorphic at $u = 0$, and*

$$\boxed{\text{FP} \int_0^1 \gamma_m(x) \text{Log } R_z(x) dx = m![u^m] \left\{ \Gamma(u)\sigma^{-u}(\text{Li}_{1+u}(q) - \zeta(1 + u)) + \frac{\text{Log } C}{u} + \frac{1}{u^2} \right\}}. \quad (6.27)$$

In particular, if a superscript $[1]$ denotes differentiation of the polylogarithm with respect to its order, then

$$\begin{aligned} \text{FP} \int_0^1 \gamma_0(x) \text{Log } R_z(x) dx &= \gamma_1 + \frac{\gamma^2}{2} - \frac{\zeta(2)}{2} - \frac{(\text{Log } \sigma)^2}{2} \\ &\quad + \text{Li}_1^{[1]}(q) + (\gamma + \text{Log } \sigma) \text{Log}(1 - q). \end{aligned} \quad (6.28)$$

Proof. Differentiate (6.20) at $v = 0$ with $p = L = 0$, using (6.26). The derivative of $-1/(u + v)$ is $1/u^2$. Expansion of

$$\Gamma(u)\sigma^{-u} = \frac{1}{u} \exp \left(-Au + \sum_{j \geq 2} \frac{(-1)^j \zeta(j)}{j} u^j \right), \quad A = \gamma + \text{Log } \sigma,$$

and $\zeta(1 + u) = u^{-1} + \gamma - \gamma_1 u + O(u^2)$ proves (6.28). The coefficient of u^{-1} cancels because $\text{Li}_1(q) = -\text{Log}(1 - q)$ and $\text{Log } C = \text{Log}(1 - q) - \text{Log } \sigma$. $\square$

At $z = \epsilon i\pi$, the polylogarithm terms disappear and the identity becomes the compact evaluation

$$\text{FP} \int_0^1 \gamma_0(x) \left\{ \log \frac{\sin \pi x}{\pi} + \epsilon \pi i x \right\} dx = \gamma_1 + \frac{\gamma^2}{2} + \frac{\pi^2}{24} - \frac{\log^2(2\pi)}{2} + \frac{\epsilon \pi i \log(2\pi)}{2}. \quad (6.29)$$

The imaginary part is also an ordinary consequence of Raabe's log-Gamma integral. The constant real part supplies a useful independent check of the branch and finite-part conventions. The $q = 0$ specialization is compatible with the prior sine-power completion; the arbitrary nonreal z identity is the new extension.

6.6 Exact derivatives and normalized primitives in the resolvent parameter

Denote the left side of (6.21) by $M_{m,p;L,k}(z)$. Coefficientwise differentiation in z is legitimate by Theorem 6.4. Since $\partial_z \text{Log } R_z = R_z$, we obtain the exact ladder

$$\frac{d}{dz} M_{m,p;L,k}(z) = L M_{m,p;L+1,k}(z) + k M_{m,p;L+1,k-1}(z), \quad (6.30)$$

with the last term omitted for $k = 0$. This differentiates the actual fixed-coordinate moments, rather than an uncompleted spectral limit.

For $L \neq 1$ define

$$P_{m,p;L,k}(z) = \sum_{j=0}^k \frac{(-1)^j k!}{(k-j)!(L-1)^{j+1}} M_{m,p;L-1,k-j}(z). \quad (6.31)$$

For $L = 1$ put $P_{m,p;1,k} = M_{m,p;0,k+1}/(k+1)$. Equation (6.30) telescopes to $P' = M$. Consequently

$$\int_{z_0}^z M_{m,p;L,k}(\xi) d\xi = P_{m,p;L,k}(z) - P_{m,p;L,k}(z_0) \quad (6.32)$$

for any path within the same open half-plane. This fixes every additive constant explicitly. In particular (6.28) is an explicit primitive of the ordinary γ_0 resolvent integral.

An exact boundary counterexample to naive substitution. The meromorphic transform is not, at every exceptional point, the pointwise integral of the specialized integrand. At $p = 2$, $u = 1$, $\lambda = 1$, the latter is zero because $\partial_x^2 \zeta(0, x) = 0$. Continuing (6.11) along $\lambda = 1$ first gives -1 . Indeed $\mathcal{L}_1(u-2, q) = (q-1) \text{Li}_{u-2}(q)/q$, with its removable $q = 0$ interpretation, gives -1 at $u = 1$ by elementary algebra. The sufficient convergence domain has $\Re(u + \lambda) > p$ and excludes this boundary. Locally the discrepancy is the limit of $(u-1)(u-2)/(u-1) = u-2$. This example identifies a concrete normalization error in an uncompleted extension; it is not an allegation that the inspected incoming resolvent theorem makes that error.

7 Exact ordinary moments of continuous resolvent powers

The continuous-power transform has a useful specialization that remains an ordinary convergent integral throughout its stated domain. It gives every polynomial moment from a regularized Gauss function, and at the two distinguished imaginary parameters the answer becomes a Gamma quotient. Generalized harmonic numbers then organize all Bernoulli moments at once.

For $z \in \mathbb{C} \setminus \mathbb{R}$, retain

$$\varepsilon = \text{sgn } \text{Im } z, \quad C = -\frac{1}{z + \varepsilon \pi i}, \quad q = \frac{z - \varepsilon \pi i}{z + \varepsilon \pi i}, \quad \sigma = -2\varepsilon \pi i.$$

Thus $|q| < 1$. On $0 < x < 1$ put

$$R_z(x) = \frac{1}{\pi \cot(\pi x) - z}, \quad R_z(x)^\lambda = \exp(\lambda \text{Log } R_z(x)),$$

where the principal logarithm is continuous along the interval. Its argument runs from 0 to $\varepsilon \pi$. Powers of C use its principal logarithm as well. In particular,

$$R_z(x)^\lambda = C^\lambda \left(\frac{1 - e^{-\sigma x}}{1 - q e^{-\sigma x}} \right)^\lambda,$$

where the power in parentheses is the radial boundary value of the holomorphic power that equals one at the origin of the disk.

We use Olver's regularized Gauss function

$$\mathbf{F}(a, b; c; q) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{\Gamma(c+n)n!} q^n. \quad (7.1)$$

When c is not a nonpositive integer, this equals ${}_2F_1(a, b; c; q)/\Gamma(c)$. The series definition is essential at apparent parameter resonances.

Theorem 7.1 (Exponential moment generator). *For $\operatorname{Re} \lambda > -1$ and every $t \in \mathbb{C}$, set $\xi = t/\sigma$. Then*

$$\int_0^1 e^{tx} R_z(x)^\lambda dx = C^\lambda e^{t/2} \frac{\Gamma(1+\lambda)}{\Gamma(1+\xi)} \mathbf{F}(\lambda, -\xi; 1+\lambda-\xi; q). \quad (7.2)$$

Both sides are entire functions of t . They are holomorphic in λ throughout the half-plane $\operatorname{Re} \lambda > -1$.

Proof. The endpoint expansions are

$$R_z(x) = x(1 + O(x)), \quad R_z(1-x) = -x(1 + O(x)).$$

The prescribed branches therefore bound the integrand, uniformly on compact parameter sets, by an integrable constant times $[x(1-x)]^\alpha$, with $\alpha > -1$. Parameter derivatives add only powers of endpoint logarithms. This proves convergence and the stated parameter holomorphy for the integral.

Write

$$f(w) = \left(\frac{1-w}{1-qw} \right)^\lambda = \sum_{n \geq 0} d_n w^n, \quad d_0 = 1.$$

First suppose $\operatorname{Re} \lambda > 0$ and $\operatorname{Re} \xi < 0$. The coefficients are absolutely summable: the coefficients of $(1-w)^\lambda$ are $O(n^{-1-\operatorname{Re} \lambda})$, with the terminating cases understood separately, while $(1-qw)^{-\lambda}$ has geometrically decaying coefficients. Consequently termwise integration gives

$$C^{-\lambda} \int_0^1 e^{tx} R_z(x)^\lambda dx = (e^t - 1) \sum_{n \geq 0} \frac{d_n}{t - \sigma n} = -\frac{e^t - 1}{\sigma} \int_0^1 u^{-\xi-1} (1-u)^\lambda (1-qu)^{-\lambda} du.$$

Euler's integral for the regularized Gauss function [8, 15.6.1] evaluates the last expression as

$$\frac{e^t - 1}{t} \Gamma(1-\xi) \Gamma(1+\lambda) \mathbf{F}(\lambda, -\xi; 1+\lambda-\xi; q).$$

The Gamma reflection formula and $\sigma = -2\varepsilon\pi i$ give

$$\frac{e^t - 1}{t} \Gamma(1-\xi) = \frac{e^{t/2}}{\Gamma(1+\xi)}.$$

This proves (7.2) on the initial open domain.

For $|q| < 1$, the series (7.1) converges normally on compact subsets of its parameters and is entire in its three parameters. The right side of (7.2) is therefore entire in t and holomorphic for $\operatorname{Re} \lambda > -1$. The identity theorem first extends in t and then in λ , proving the full statement. In particular, no exclusion of $1+\lambda-\xi = 0, -1, -2, \dots$ is necessary in the regularized notation. $\square$

The ingredients in this proof are classical Beta and Gauss identities. The contribution here is their branch-consistent specialization to the continuous-power resolvent and its exact moment consequences.

Corollary 7.2 (Means and logarithmic moments). *For $\operatorname{Re} \lambda > -1$ and $k \geq 0$,*

$$\int_0^1 R_z(x)^\lambda (\operatorname{Log} R_z(x))^k dx = C^\lambda (\operatorname{Log} C)^k. \quad (7.3)$$

In particular, $\int_0^1 R_z(x)^\lambda dx = C^\lambda$ and $\int_0^1 (\operatorname{Log} R_z(x))^k dx = (\operatorname{Log} C)^k$.

Proof. Put $t = 0$ in (7.2), using $\mathbf{F}(\lambda, 0; 1 + \lambda; q) = 1/\Gamma(1 + \lambda)$. Differentiate k times in λ . The endpoint majorants in the proof of the theorem justify each differentiation under the integral. $\square$

7.1 The Bernoulli generator and its Gamma specialization

Let $B_m(x)$ be the Bernoulli polynomials with $B_1(x) = x - 1/2$. Multiplication of (7.2) by $t/(e^t - 1)$ gives the exact generator

$$\begin{aligned} \sum_{m \geq 0} \frac{t^m}{m!} \int_0^1 B_m(x) R_z(x)^\lambda dx &= C^\lambda \Gamma(1 - t/\sigma) \Gamma(1 + \lambda) \\ &\times \mathbf{F}(\lambda, -t/\sigma; 1 + \lambda - t/\sigma; q), \quad |t| < 2\pi. \end{aligned} \quad (7.4)$$

The Bernoulli series converges uniformly in $x \in [0, 1]$ on every smaller t -disk. The same integrable endpoint majorant justifies the interchange with the integral. Thus coefficient extraction is an ordinary-integral identity, including when $\lambda = -1/2$.

Corollary 7.3 (A digamma–Lerch first moment). *For $\operatorname{Re} \lambda > -1$,*

$$\begin{aligned} \int_0^1 \left(x - \frac{1}{2}\right) R_z(x)^\lambda dx &= \frac{C^\lambda}{\sigma} [\psi(1 + \lambda) + \gamma + q\Phi(q, 1, 1 + \lambda) \\ &+ \operatorname{Log}(1 - q)], \end{aligned} \quad (7.5)$$

where $\Phi(q, 1, a) = \sum_{n \geq 0} q^n/(n + a)$.

Proof. With $f(w)$ as above, its nonconstant Fourier coefficients give

$$C^{-\lambda} \int_0^1 \left(x - \frac{1}{2}\right) R_z(x)^\lambda dx = -\frac{1}{\sigma} \int_0^1 \frac{f(v) - 1}{v} dv.$$

This follows first for $\operatorname{Re} \lambda > 0$ by absolute summation and then for $\operatorname{Re} \lambda > -1$ by holomorphy. For $0 < q < 1$, substitute $u = (1 - v)/(1 - qv)$ in the last integral:

$$\int_0^1 \frac{f(v) - 1}{v} dv = \int_0^1 (u^\lambda - 1) \left(\frac{1}{1 - u} - \frac{q}{1 - qu} \right) du = -\psi(1 + \lambda) - \gamma - q\Phi(q, 1, 1 + \lambda) - \log(1 - q).$$

The first equality extends to $|q| < 1$ by holomorphy. Substitution proves (7.5) with the principal logarithm. $\square$

Define complex generalized harmonic numbers on $\operatorname{Re} \lambda > -1$ by

$$H_\lambda^{(1)} = \psi(1 + \lambda) + \gamma, \quad H_\lambda^{(j)} = \zeta(j) - \zeta(j, 1 + \lambda) \quad (j \geq 2). \quad (7.6)$$

For nonnegative integers $\lambda = N$ these are the usual finite harmonic numbers. Define $h_m(\lambda)$ by

$$\sum_{m \geq 0} h_m(\lambda) u^m = \frac{\Gamma(1 - u) \Gamma(1 + \lambda)}{\Gamma(1 + \lambda - u)} = \exp \left(\sum_{j \geq 1} \frac{H_\lambda^{(j)}}{j} u^j \right) \quad (|u| < \min\{1, |1 + \lambda|\}). \quad (7.7)$$

The Taylor series of the Gamma quotient itself is holomorphic for $|u| < 1$. The smaller disk in the exponential representation accounts for possible zeros of that quotient when λ has negative real part; its logarithmic series must not be continued through such a zero. Equivalently,

$$h_0 = 1, \quad h_m = \frac{1}{m} \sum_{j=1}^m H_\lambda^{(j)} h_{m-j} \quad (m \geq 1). \quad (7.8)$$

These are complete homogeneous polynomials in the generalized harmonic coordinates; for example,

$$h_1 = H_\lambda^{(1)}, \quad h_2 = \frac{(H_\lambda^{(1)})^2 + H_\lambda^{(2)}}{2}, \quad h_3 = \frac{(H_\lambda^{(1)})^3 + 3H_\lambda^{(1)}H_\lambda^{(2)} + 2H_\lambda^{(3)}}{6}.$$

Theorem 7.4 (Every Bernoulli moment at the imaginary base points). *For $z = \varepsilon\pi i$, $\operatorname{Re} \lambda > -1$, and every integer $m \geq 0$,*

$$\int_0^1 B_m(x) \left(\frac{\sin(\pi x)}{\pi} e^{\varepsilon\pi i x} \right)^\lambda dx = m! \left(\frac{\varepsilon i}{2\pi} \right)^\lambda (-2\varepsilon\pi i)^{-m} h_m(\lambda). \quad (7.9)$$

The power on the left means $\exp\{\lambda[\log(\sin(\pi x)/\pi) + \varepsilon\pi i x]\}$.

Proof. At these points $q = 0$ and $C = \varepsilon i/(2\pi)$, while $R_{\varepsilon\pi i}(x) = \sin(\pi x)e^{\varepsilon\pi i x}/\pi$. Equation (7.4) therefore reduces to C^λ times the Gamma quotient (7.7) at $u = t/\sigma$. Comparing coefficients proves the formula. The logarithmic Taylor expansion of that quotient gives (7.7) from (7.6); logarithmic differentiation then gives (7.8). $\square$

At the negative half power,

$$H_{-1/2}^{(1)} = -2 \log 2, \quad H_{-1/2}^{(j)} = -(2^j - 2)\zeta(j) \quad (j \geq 2).$$

The first two nonconstant moments are consequently

$$\begin{aligned} \int_0^1 \left(x - \frac{1}{2} \right) \sqrt{\frac{\pi}{\sin(\pi x)}} e^{-\varepsilon\pi i x/2} dx &= -\frac{1 + \varepsilon i}{\sqrt{\pi}} \log 2, \\ \int_0^1 \left(x^2 - x + \frac{1}{6} \right) \sqrt{\frac{\pi}{\sin(\pi x)}} e^{-\varepsilon\pi i x/2} dx &= \sqrt{\pi}(1 - \varepsilon i) \left(\frac{1}{12} - \frac{\log^2 2}{\pi^2} \right). \end{aligned} \quad (7.10)$$

Both endpoints have only inverse-square-root singularities, so these are absolutely convergent ordinary integrals. For example the second identity also yields the entirely real evaluation

$$\int_0^1 \left(x^2 - x + \frac{1}{6} \right) \frac{\cos(\pi x/2)}{\sqrt{\sin(\pi x)}} dx = \frac{1}{12} - \frac{\log^2 2}{\pi^2}. \quad (7.11)$$

At the positive half power, $H_{1/2}^{(1)} = 2 - 2 \log 2$ and $H_{1/2}^{(j)} = 2^j - (2^j - 2)\zeta(j)$ for $j \geq 2$. Thus every moment at either half power belongs to an explicitly specified algebra of zeta values, $\log 2$, π , and the displayed square-root prefactor. Further λ -derivatives of (7.9) give all corresponding moments with arbitrary powers of $\operatorname{Log} R_{\varepsilon\pi i}$.

Verification. The independent script `check_moments.py` compares the Gamma–Gauss generator against direct real-interval quadrature in both half-planes, including nonintegral powers with negative real part and powers with real part greater than one. It also compares four Bernoulli moments with (7.9), reconstructs the first nine Fourier coefficients by two exact symbolic formulas, and rejects a changed coefficient. Its separate spectral and finite-part checks audit the conventions used in the surrounding continuous-power transform. These numerical comparisons are diagnostics, not interval certificates or steps in the proofs above.

8 Source audit and proposed integration

8.1 What the audit established

No new false theorem was found in the specific source arguments used for the present results. In particular, the incoming cubic bridge, the centered parity criterion, and the previously proved trigamma-square moments survive the checks relevant to their use here. Their stated limits are research questions, not flaws in their proofs. This is a targeted mathematical audit of the relevant material, not a certification of every page in the repository.

The appropriate source changes are therefore progress updates and additional normalization examples. Table 1 gives the precise mapping. The supplied integration notes identify the archive-internal source paths and theorem labels; no source repository was modified in preparing this package.

Source question	Result established here	Remaining scope
Cubic harmonic report, Q6	Quartic power-to-harmonic reduction, Theorem 3.1	Powers $k \geq 5$ and arithmetic reduction of the retained depth-three coefficient
Harmonic-parity report, R1	All pure two-even-tail moments and their generator, Theorems 4.2, 5.1	Additional harmonic factors and products of four or more tails
Harmonic-parity report, R4	Complex powers, joint resonant completion, all logarithmic moments, Theorem 6.4	Products of distinct resolvents or nonlinear continuous-order factors
Higher-reflection report, centered products	First mixed pair and all two-tail orders	Higher-depth products and spectral derivatives
Canonical Gaussian S_6 , revised S_8	No proof or refutation obtained	Exact functional bridge to the frozen candidates

Table 1: Scoped updates relative to the pinned source revision.

8.2 A concrete correction to a tempting extension

There is a mathematically incorrect extension that the new results rule out: replacing a specialized coordinate finite part by the meromorphic transform at every point where the specialized integrand happens to be integrable. The counterexample at the end of Section 6 is exact. With $p = 2$, $u = 1$, and $\lambda = 1$, the ordinary integral of $\partial_x^2 \zeta(0, x) R_z(x)$ equals zero. The fixed- λ meromorphic continuation of the spectral formula equals -1 . The local monomial has a coefficient that vanishes at the same point as its integration denominator, leaving the nonzero limit $u - 2$.

The correction is to retain the entire moving amplitude before coefficient extraction, as in (6.20). Removing just the negative Laurent coefficients of a one-variable expansion is insufficient. At negative integer powers the two smooth endpoint residues cancel, yet their quotient has the nonzero limit (6.23). A cancellation of residues therefore does not imply that there is no finite endpoint correction.

This is a warning and an exact regression example for future extensions. The inspected incoming resolvent theorem states its own convergence and completion conditions; this article does not attribute the incorrect extension to that theorem. The package includes a deliberately corrupted coefficient check so that omitting the endpoint phase is visibly rejected by exact algebra.

8.3 Four conventions that should accompany integration

1. **Keep strict harmonic coordinates named and normalized.** The quartic coefficient θ is the coefficient of u in $Z_3(1 + u, 1, 1)$, not the constant coefficient and not an order derivative of a regular function before its poles have been specified. The entire principal part and constant coefficient are given in (3.2). The retained coefficient is genuine remaining arithmetic data in the present reduction; its integral representation is a normalization, not an ordinary- constant evaluation.
2. **Retain subtractions inside convergent summands.** The sums in (4.4) and (4.27) are ordinary convergent sums of the displayed brackets. Their polynomial pieces must not be separated and summed as ordinary series. The equality with the continued Dirichlet series is proved using the half-shift and Hurwitz trivial zeros.
3. **Use the regularized Gauss function at parameter resonances.** The definition (7.1) remains valid when its third parameter is a nonpositive integer. A quotient written informally as ${}_2F_1/\Gamma(c)$ must be read through that definition. Likewise the logarithmic Gamma-quotient series has the smaller disk stated in (7.7); zeros of the quotient obstruct its logarithm before they obstruct the quotient itself.
4. **Distinguish analytic proofs, finite algebra, and numerical evidence.** The infinite identities follow from the proofs. Symbolic checks verify selected finite polynomial consequences. The numerical files use ordinary multiprecision arithmetic with specified analytic tail estimates. They are not interval-enclosed evaluations and do not certify arithmetic independence of any constants.

8.4 The Gaussian conjectures and the exact obstruction

The frozen S_6 target is in `chapters/04-cyclotomic-quotients.tex`; the revised S_8 candidate is in `chapters/04-S8-candidate.tex` of the manuscript at the pinned revision [1]. Both remain conjectures. Their existing short candidate formulas have not been replaced by a numerical fit in this work.

The exact-resonant report already proves a specific obstruction: the S_6 target is separated from the entire convergent Cayley shuffle ideal together with a specified family of 5,131 additional relations [6]. This is stronger than an unsuccessful small linear solve. It is also a statement about that named formal relation system, not a proof that the numerical period conjecture is false or that no other functional relation can prove it.

The larger search in the independent-orders report generated 51,244 double-shuffle rows in 24,514 imaginary word coordinates. Its modular elimination stopped after 8,500 pivots, before exhausting the system [7]. The incomplete search cannot be cited as an exhausted obstruction, and it supplies no rational identity certificate. No new claim about its terminal rank is made here.

The new complex-power transform provides analytic identities that can be tested against a Gaussian target, but this article has not proved a specialization producing either frozen residual. The unresolved step is an explicit functional bridge, followed by exact coefficient extraction. Giving another representation of an unrelated zeta constant does not accomplish that step.

9 Further research questions and an order of attack

The following questions ask for exact identities, finite reductions, or normalized analytic representations. They are not asserted conjectures unless an implication is explicitly proved above.

The requested deliverable for each question is stated to separate mathematical progress from additional numerical evidence.

9.1 Higher digamma powers and harmonic Laurent coefficients

1. The fifth power with all finite corrections. Repeat the quartic contour and finite summation argument for $\text{FP} \int_0^1 \psi(x)^5 dx$. The principal parts are algorithmically finite harmonic polynomials, but their integration requires a new exact summation-by-parts reduction. Determine which linear Laurent coefficients of strict depth-four and lower-depth zeta functions enter, and give every ordinary zeta derivative and Stieltjes correction. A proof should include the constant term of each finite asymptotic sum, not only matching principal parts or a high-precision decimal.

2. Canonical polynomials that remove lower-depth data. The polynomial $T^4 + 4\gamma T^3$ removes η from the quartic bridge. Is there a systematically constructed monic polynomial $P_k(T)$, with coefficients in a stated ordinary-constant algebra, for which $\text{FP} \int_0^1 P_k(\psi(x)) dx$ retains only one new highest-depth harmonic coefficient? The Gamma and inverse-Gamma coefficient polynomials are natural candidates for organizing the recursion. A useful theorem would give the recurrence and prove the cancellation for arbitrary k ; the single quartic cancellation does not establish such a theorem.

3. The arithmetic content of the depth-three coordinate. The normalized elementary kernel (3.22) determines θ exactly. Seek an independent identity placing it in a smaller algebra of zeta derivatives, ordinary Stieltjes constants, or Gamma/Barnes integrals. The target should state which coordinates are eliminated. Merely replacing the strict Laurent definition by another name for the same integral does not reduce the arithmetic content. Conversely, a failed numerical relation search does not prove irreducibility.

4. A common-Hurwitz-shift quartic bridge. For $a > 0$, define a consistent common shift in every denominator of the strict depth-three series and derive the counterpart of (3.20). Determine the associated identity for products of $\psi(a+x)$ and its derivatives, including the two ordinary endpoint values on $[a, a+1]$. Then analyze the singular limit $a \downarrow 0$ with a specified coordinate finite part. Replacing γ_m by $\gamma_m(a)$ in the unshifted formula without deriving these endpoint terms is not valid evidence.

9.2 Harmonic-tail arithmetic beyond two factors

5. The remaining harmonic factors in R1. For the products

$$A_p(n)A_q(n)H_n^a \prod_{j \geq 1} (H_n^{(2j+1)})^{b_j},$$

with finitely many nonzero b_j , determine finite formulas for the regular centered values and a moment-independent coordinate space for each fixed polynomial. The source parity theorem predicts local regularity, but it does not evaluate the global finite constant. Finite summation by parts increases the depth of the harmonic sums. One must specify the necessary multiple-zeta coordinates and prove any claimed depth reduction.

6. Four even tails and the first genuine depth increase. Carry out the analogous calculation for $A_p A_q A_r A_s$, first for A_2^4 and $A_2^3 A_4$. An even number of even tails preserves the centered even-power asymptotics. The two-tail theorem suggests a finite Bernoulli reduction at every moment order, but higher-depth convergent sums can survive. Determine a fixed spanning

set and an analytic generating function. A parity argument alone does not prove that all such higher-depth values reduce to ordinary zeta products.

7. First spectral derivatives at the regular lattice. Find exact reductions of $D'_{p,q}(-2r)$ or combinations of these derivatives. Differentiating the subtracted continuation produces logarithmically weighted global remainder sums as well as derivatives of the explicitly removed Hurwitz terms. Seek identities of these kernels that cancel the global terms, or identify a minimal additional harmonic-Stieltjes coordinate family. The finite value formula cannot simply be differentiated with respect to its discrete integer label r .

8. Finite generation by initial moments. For fixed even p, q , Corollary 4.3 gives at most $\max(p, q)$ specified arithmetic coordinates. Determine whether a uniform finite prefix of the moment sequence generates the same rational span, and exhibit a rational inverse matrix when it does. The $(2, 4)$ case is proved by its first four moments. An optimal bound concerns the rank of a specified coefficient matrix. Equality with a dimension of actual constants would require an additional arithmetic independence result.

9. Rational shifts and colored polylogarithms. Replace the inner shift $n + 1$ and the center $n + 1/2$ by compatible rational shifts. Classify when the removed polynomial sums vanish, and determine the surviving finite corrections when they do not. The natural targets are Hurwitz zeta values at rational arguments and colored Euler sums, together with polylogarithms at roots of unity. A paired formulation at shifts a and $1 - a$ may retain more symmetry than an isolated noncentral shift.

9.3 Resolvent products, spectral derivatives, and primitives

10. Several distinct nonreal resolvents. For

$$\prod_{j=1}^d R_{z_j}(x)^{\lambda_j},$$

derive an explicit joint Fourier or beta transform and complete the hyperplanes where the total endpoint exponent is integral. When all z_j lie in the same half-plane the disk factors have a common Fourier orientation; an Appell or Lauricella function is a plausible ordinary-moment generator. Parameters in opposite half-planes introduce two-sided convolutions and additional arithmetic structure. Each claim needs a branch prescription and the full endpoint amplitude before any specialization.

11. Products of continuous polygamma orders. The transform in this article is linear in the Hurwitz factor. For products of two independently varying Hurwitz orders, derive the convolution and identify exactly when ordinary polylogarithm order derivatives suffice and when ordered or Tornheim coordinates remain. Mixed order derivatives should be extracted from a joint completed germ. Multiplying two separately completed scalar identities does not account for the product's local singularities.

12. Arithmetic at rational complex-power exponents. At $q = 0$, every Bernoulli moment is a finite polynomial in the generalized harmonic values $H_\lambda^{(j)}$ and the fixed prefactor. At rational λ these values have classical Hurwitz-zeta descriptions. Determine comparable finite reductions for useful nonzero algebraic q and rational λ , especially for derivatives in λ and the polylogarithm order. The beta-polylogarithm formula supplies a convergent starting integral; it does not by itself prove a finite constant evaluation.

13. Normalized primitives beyond the balanced combinations. Combine the explicit z -ladder with argument integration and Hurwitz order differentiation to seek individual Barnes-type representations. State the additive normalization at a base point and derive its transport under z and argument shifts. The existing difference primitives already fix their constants; a new result should identify a smaller named function class or an additional functional equation, rather than introduce an arbitrary integration constant.

9.4 The Gaussian bridge and exact certification

14. A functional identity for the frozen Gaussian residual. Express the actual S_6 or revised S_8 residual as a coefficient of a proved joint generating identity, with all changes of variables and branches explicit. The Cayley-ideal obstruction specifies information that its named relation family cannot supply. A successful new relation should have a computable nonzero action on the separating functional, or an analytic specialization demonstrably outside that family. The article's resolvent transform is a source of candidate relations, not an already established bridge.

15. Complete and certify any enlarged relation search. If the larger modular calculation is resumed, first preserve the frozen target, relation enumeration, and checkpoint hashes. Finish the declared relation system and recover an exact rational certificate or exact separating functional. A modular rank observation alone is not a rational proof of the target identity, and an interrupted elimination does not show nonmembership in the generated span. The input system and the terminal conclusion must be independently replayable.

16. Formalize reusable analytic and algebraic components. The most reusable formal units are the two-point Mittag-Leffler subtraction, the finite Faulhaber constant-term calculation, the beta coefficient identity (6.9), and the moving-monomial completion lemma inside Theorem 6.4. Formalizing these would make subsequent all-index identities considerably easier to verify. An interval evaluation of θ or Q_4 can be added separately by enclosing the same analytic tail formulas; it would strengthen the numerical artifact, not replace the exact proof.

9.5 Practical sequence

The most direct next calculations are the fifth-power reduction, the initial-moment coefficient matrices, and the first additional harmonic factors in the two-tail family. Each has a finite exact starting point supplied here. Four-tail products and joint resolvent powers are natural next structural extensions. The Gaussian reduction should proceed only with a concrete candidate functional bridge or a completed exact relation computation; more precision for the same unproved scalar fit does not remove the known missing step.

A Verification of the mixed-tail identities

The script `verification/verify_mixed_tails.py` records finite exact checks and a numerical route independent of the moment formula. Its numerical tail approximation has an explicit analytic remainder estimate. Ordinary multiprecision arithmetic is used, so the recorded floating-point computations are not asserted to be interval certificates. The identities are established by the proofs above.

A.1 A proved truncation envelope

For real $x > 0$, put

$$T_p(x) = \zeta(p, x + 1/2), \quad Q_{p,K}(x) = \sum_{j=0}^K c_{p,j} x^{1-p-2j}, \quad R_{p,K} = \frac{2(p+2K)!}{(p-1)!(2\pi)^{2K+2}}.$$

Then

$$0 < T_p(x) \leq \frac{x^{1-p}}{p-1}, \quad |T_p(x) - Q_{p,K}(x)| \leq R_{p,K} x^{-p-2K-1}. \quad (\text{A.1})$$

Here and below the bound is needed only to substantiate numerical verification, not as the research objective.

To prove it, let $g(u) = u/(2 \sinh(u/2))$. The standard hyperbolic-cosecant partial fractions [8, Eq. 4.36.5] give

$$g(u) = 1 + 2 \sum_{m=1}^{\infty} (-1)^m \frac{u^2}{u^2 + (2\pi m)^2}.$$

Expanding each rational summand through degree $2K$, and using the alternating-series estimate on the remainder, gives for real $u > 0$

$$\left| g(u) - \sum_{j=0}^K \frac{B_{2j}(1/2)}{(2j)!} u^{2j} \right| \leq \frac{2u^{2K+2}}{(2\pi)^{2K+2}}.$$

The identity

$$T_p(x) = \frac{1}{(p-1)!} \int_0^{\infty} u^{p-2} e^{-xu} g(u) \, du$$

now proves both parts of (A.1); the positivity bound uses $0 < g(u) \leq 1$.

Fix $N \geq 1$, put $a = N + 1/2$, and choose $K \geq K_r$. Define

$$M_{p,K}(a) = \sum_{j=0}^K |c_{p,j}| a^{-2j}, \quad k_{p,q,\ell}^{[K]} = \sum_{\substack{0 \leq j \leq K \\ 0 \leq \ell-j \leq K}} c_{p,j} c_{q,\ell-j}.$$

The finite computational expression is

$$\begin{aligned} V_{p,q,r}^{N,K} &= \sum_{n=0}^{N-1} \left\{ x_n^{2r} A_p(n) A_q(n) - \sum_{\ell=0}^{K_r-1} k_{p,q,\ell} x_n^{2r+2-W-2\ell} \right\} \\ &\quad + \sum_{\ell=K_r}^{2K} k_{p,q,\ell}^{[K]} \zeta(W-2+2\ell-2r, a). \end{aligned} \quad (\text{A.2})$$

All the Hurwitz-zeta orders in the last sum are at least two. Provided $W + 2K - 2r > 1$, its error satisfies

$$|D_{p,q}(-2r) - V_{p,q,r}^{N,K}| \leq \left\{ \frac{R_{p,K}}{q-1} + M_{p,K}(a) R_{q,K} \right\} \zeta(W+2K-2r, a). \quad (\text{A.3})$$

One may take the smaller bound obtained by interchanging p, q . Indeed, (A.1) gives

$$|T_p T_q - Q_{p,K} Q_{q,K}| \leq \left\{ \frac{R_{p,K}}{q-1} + M_{p,K}(a) R_{q,K} \right\} x^{-W-2K} \quad (x \geq a).$$

Multiplication by x^{2r} , summation over the omitted tail, and expansion of the finite product prove (A.3).

A.2 Recorded checks

The script uses 110 decimal working digits, $N = 64$, and $K = 30$. It checks $r = 0, 1, 2, 3, 5, 6$ for each of

$$(p, q) = (2, 2), (2, 4), (4, 4), (2, 6), (4, 6), (6, 6).$$

These include the first regularized values and several higher regularized values, not only ordinary unsubtracted sums. Every one of the 36 observed discrepancies is below its analytic truncation envelope. Selected results are

p	q	r	Absolute discrepancy	Truncation envelope
2	4	2	$1.4110 \cdot 10^{-73}$	$1.4471 \cdot 10^{-73}$
2	4	3	$5.9899 \cdot 10^{-70}$	$6.1431 \cdot 10^{-70}$
2	4	6	$4.6138 \cdot 10^{-59}$	$4.7314 \cdot 10^{-59}$
4	4	6	$7.2161 \cdot 10^{-63}$	$7.4014 \cdot 10^{-63}$
4	6	6	$1.8213 \cdot 10^{-64}$	$1.8709 \cdot 10^{-64}$
6	6	6	$5.1274 \cdot 10^{-68}$	$5.2690 \cdot 10^{-68}$

The exact checks independently expand the finite uncentered tail products and Faulhaber polynomials and extract their constant coefficients for six pairs and eleven odd block orders. They also verify the short $(2, 4)$ block formula through $j = 12$ and recover the previously proved trigamma-square formula through $r = 12$. The two eliminations in (4.26) are checked by exact rational algebra.

Finally, direct quadrature of the generator is compared with its first 41 Taylor terms for the three cases

$$(p, q, t) = (2, 4, 0.7), \quad (2, 4, 1.3), \quad (4, 6, 0.8).$$

Their respective differences are approximately $1.1 \cdot 10^{-78}$, $1.2 \cdot 10^{-56}$, and $2.6 \cdot 10^{-76}$, consistent with the omitted convergent Taylor tail. For both $(2, 4)$ checks, direct integral evaluation and the explicit Li_2 – Li_6 formula agree to better than 10^{-109} . Complete numerical values, software versions, formulas, and truncation envelopes are recorded in `verification/mixed_tail_results.json`.

B Reproducibility and verification record

The package contains the complete modular source, a self-contained single-file TeX version, the compiled PDF, four verification programs, their recorded JSON outputs, source provenance, a claim inventory, and integration notes. It does not require a checkout of ProveIt to build or run its mathematical checks. The source snapshot is recorded for comparison and attribution.

B.1 Build and replay

From the extracted package directory, run

```
python3 build.py
python3 run_verification.py
```

The first command generates the standalone TeX file and builds the PDF using pdfLaTeX. The second runs the four mathematical suites and saves a replay summary and per-suite logs under `verification/replay/`. Each suite also updates its own recorded JSON file. To preserve the shipped records verbatim, run the replay on a copy of the package.

The Python dependencies are SymPy and mpmath, with the recorded versions pinned in `requirements.txt`. The TeX build uses standard AMS, Latin Modern, geometry, microtype, hyperref, and related standard packages listed explicitly in the preamble. No external bibliographic processor, private dataset, network service, or shell-escape operation is needed.

B.2 The scope of the checks

The recorded exact checks total 183, together with 21 deliberate corruption controls. They are finite symbolic tests of separate coefficient constructions and finite algebraic consequences. They are not a count of independent theorems, and they do not replace the all-index analytic arguments.

Suite	Exact checks	Independent numerical route
Quartic bridge	44	Harmonic/Binet summation versus local Taylor integration and ordinary quadrature, at 100 working digits
Mixed tails	94	36 finite-head/Hurwitz-tail evaluations, plus generator comparisons, at 110 working digits
Continuous-power moments	9	19 quadrature, Mellin, Gamma, Lerch, and subtracted logarithmic comparisons, at 55 working digits
Endpoint contacts and primitives	36	Exact arithmetic only; 20 omitted-phase controls are rejected

The remaining corruption control changes a Fourier coefficient's prefactor. It is rejected by exact polynomial comparison. The continuous-power numerical suite samples both half-planes, complex exponents, an integrable negative half power, and nonzero complex q . Its largest absolute discrepancy is about $2.824 \cdot 10^{-31}$, arising in inverse-square-root endpoint quadrature. This is consistent with the deliberately more modest diagnostic tolerance; it is not a claim of 55 correct digits in every quadrature.

For the quartic calculation, the finest arithmetic evaluation and the independent local Taylor/quadrature evaluation differ by less than $1.40 \cdot 10^{-83}$. The Binet truncation bound is below $1.88 \cdot 10^{-83}$. For the mixed tails, every observed moment discrepancy is below its displayed analytic truncation envelope. These statements separate the analytically controlled omitted tail from floating-point rounding, which is not enclosed by this implementation. No interval certificate is claimed.

B.3 Exact contact and antiderivative checks

The fixed-coordinate correction at a negative integral power has an elementary residue interpretation. For $M \geq 1$,

$$[x^{M-1}]\{x(K(x) - z)\}^M = \operatorname{Res}_{x=0}(K(x) - z)^M dx = \frac{(-z + i\pi)^M - (-z - i\pi)^M}{2i\pi}. \quad (\text{B.1})$$

Indeed, the local coordinate $v = 1/K(x) = \tan(\pi x)/\pi$ satisfies $dx = dv/(1 + \pi^2 v^2)$ and has derivative 1 at zero. Thus the residue is the coefficient of v^{-1} in $(v^{-1} - z)^M/(1 + \pi^2 v^2)$, namely

$$\sum_{j=0}^{\lfloor (M-1)/2 \rfloor} \binom{M}{2j+1} (-z)^{M-2j-1} (-\pi^2)^j.$$

The binomial theorem gives the final expression in (B.1). Consequently the difference between the coordinate finite-part mean and its continued spectral value is

$$\frac{(-z + i\pi)^M + (-z - i\pi)^M}{2} - (-z - \epsilon i\pi)^M = \epsilon \pi i [x^{M-1}]\{x(K(x) - z)\}^M. \quad (\text{B.2})$$

This proves the contact identity for every M , independently of the general completion theorem.

The accompanying `check_contacts.py` constructs the cotangent Taylor jet from Bernoulli numbers and independently checks it against the sine and cosine jets. It then verifies (B.2)

exactly for $M = 1, \dots, 10$ and both signs of ϵ , with z and π left symbolic. Omitting the endpoint phase produces a nonzero polynomial residual in all twenty cases. In addition, the script verifies the primitive's telescoping coefficients for $k = 0, \dots, 5$ with symbolic $L \neq 1$, checks the $L = 1$ branch, and reproduces the boundary example $p = 2$, $u = \lambda = 1$: the specialized ordinary integrand is zero, the continued spectral value is -1 , and the local endpoint term has limit -1 . All thirty-six exact checks and all twenty deliberate corruption controls pass. These finite checks complement the all-index proofs; they do not replace them.

B.4 Integration and provenance artifacts

INTEGRATION.md identifies the source questions and the proposed scoped status updates. claims.json lists the principal statements, domains, proof labels, dependencies, and remaining limitations. provenance/source_manifest.json records the pinned source revision and hashes of inspected manuscript files and incoming archives. SHA256SUMS covers the deliverable files present at packaging time. The complete archive is self-contained; the original incoming ZIP files are identified by hash rather than duplicated inside it.

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